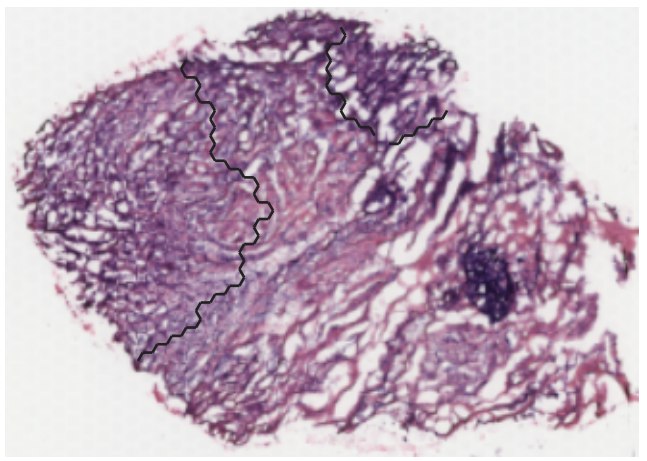
GSE171351_A1 | Tissue context

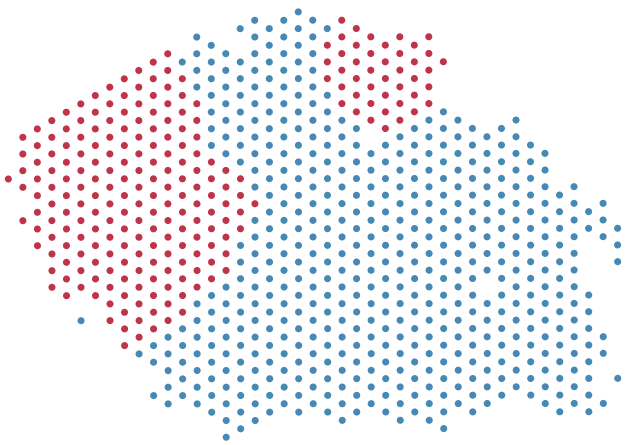
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



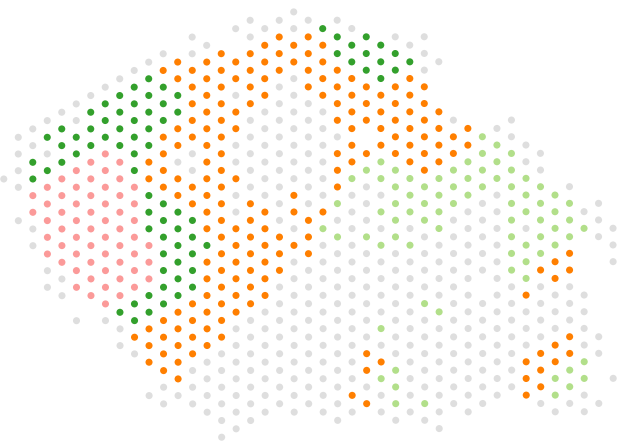
Black line: observed candidate outer edge

Final candidate location



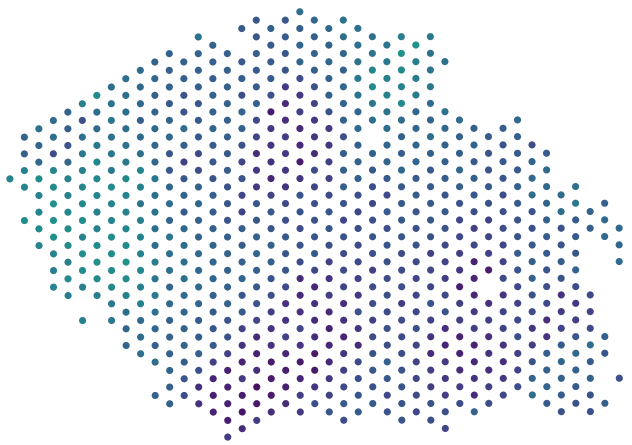
inside outside

Accepted R12 (original 0–8)



2 3 5 6

Epithelial support



0 1

Fraction of all 30

MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

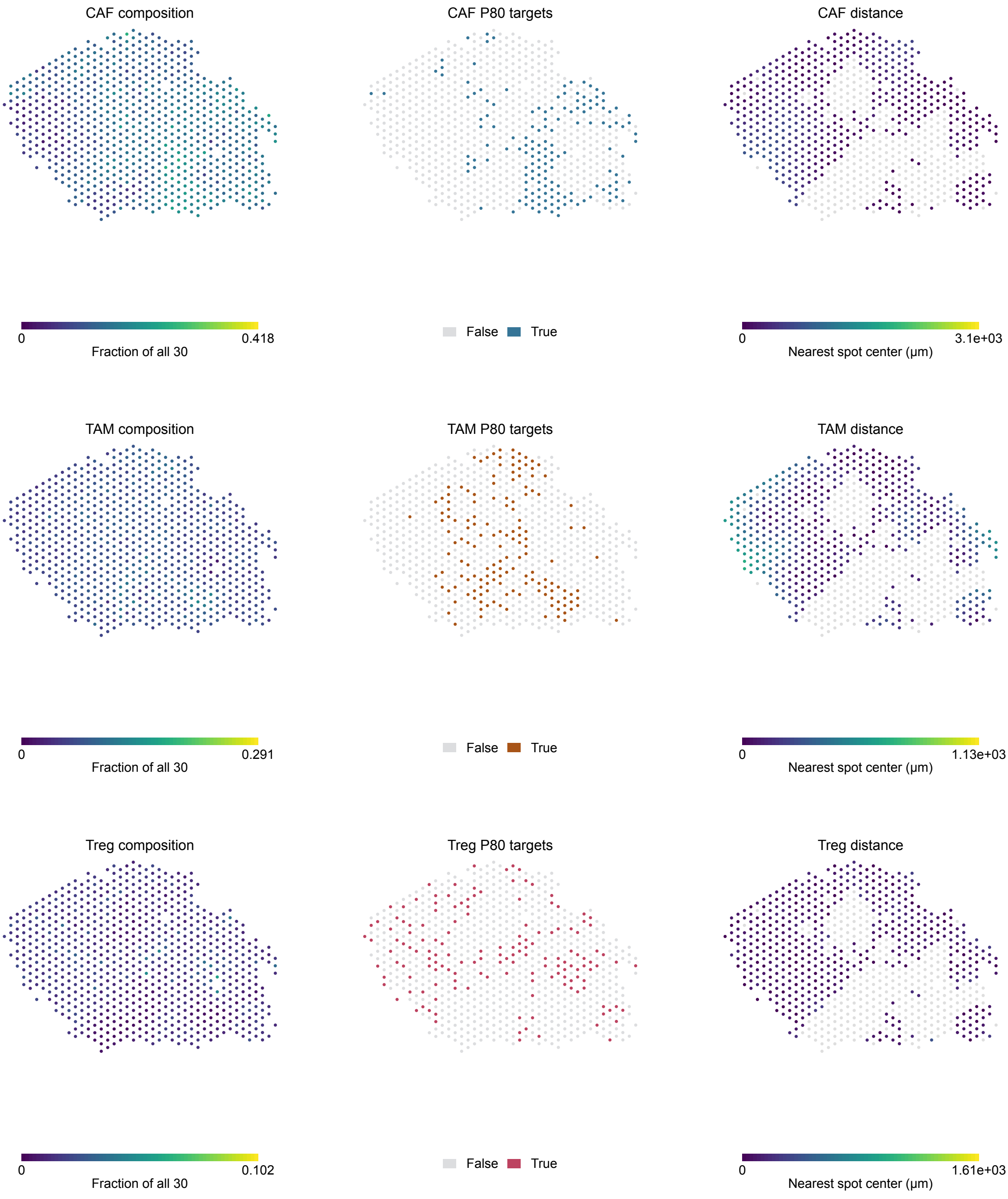


0 0.14

Matched-model AUCell

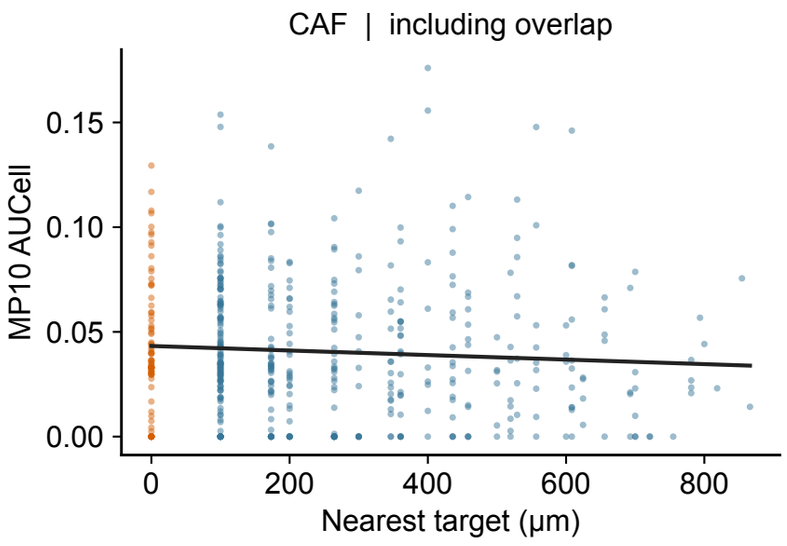
GSE171351_A1 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

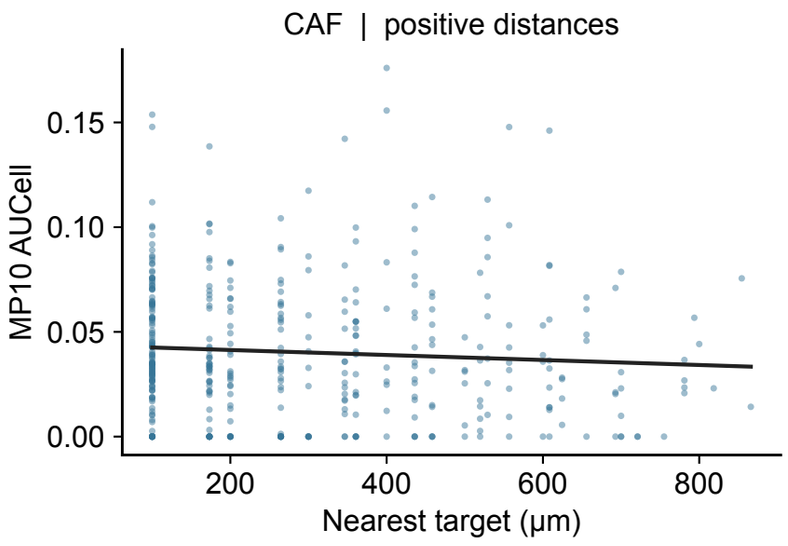


GSE171351_A1 | MP10 versus nearest target

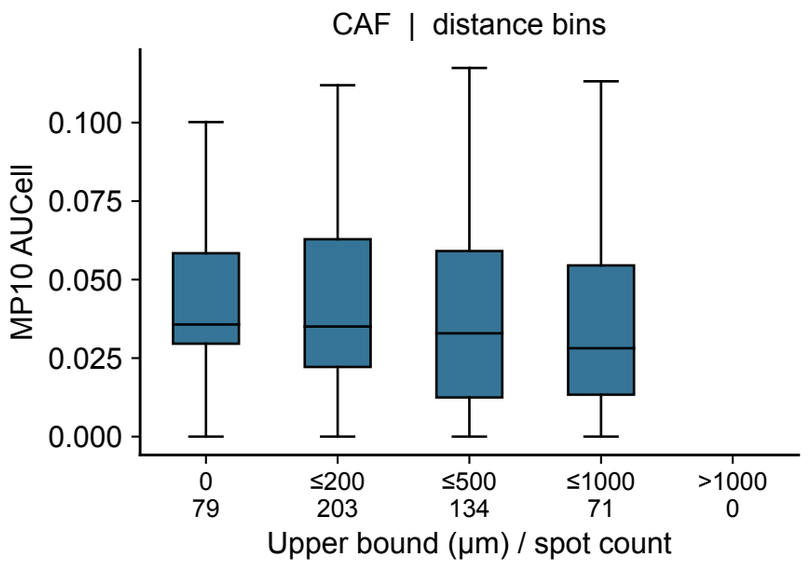
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



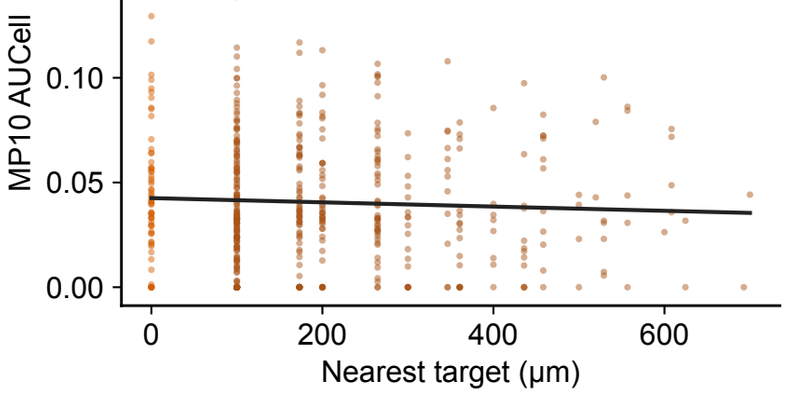
r -0.07 ρ -0.10
n 487/488 queries • 154 targets



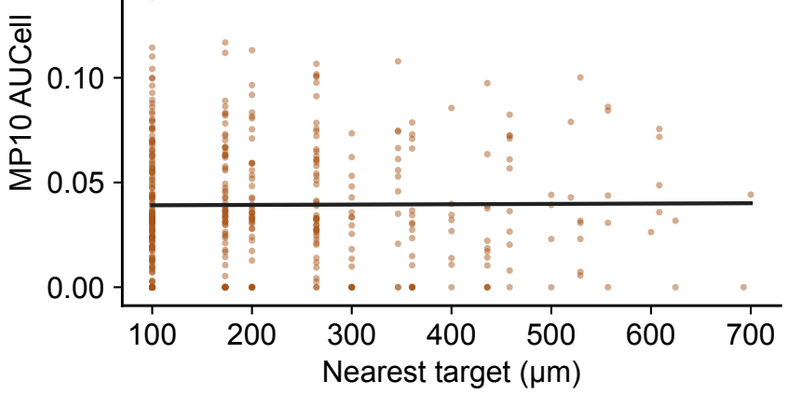
r -0.07 ρ -0.11
n 408/488 queries • 154 targets



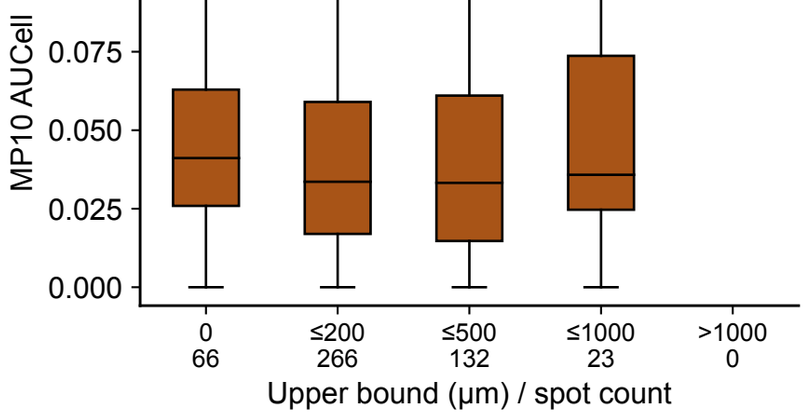
Orange = zero distance
Line = within-section linear fit



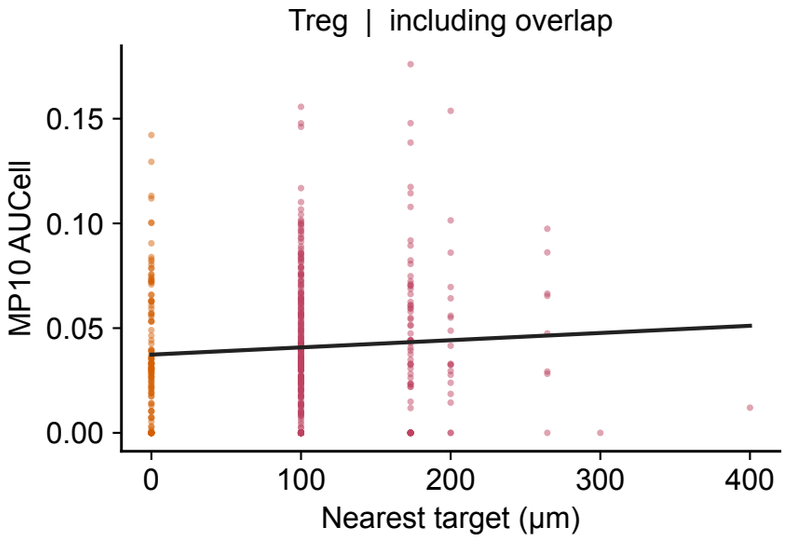
r -0.05 ρ -0.02
n 487/488 queries • 154 targets



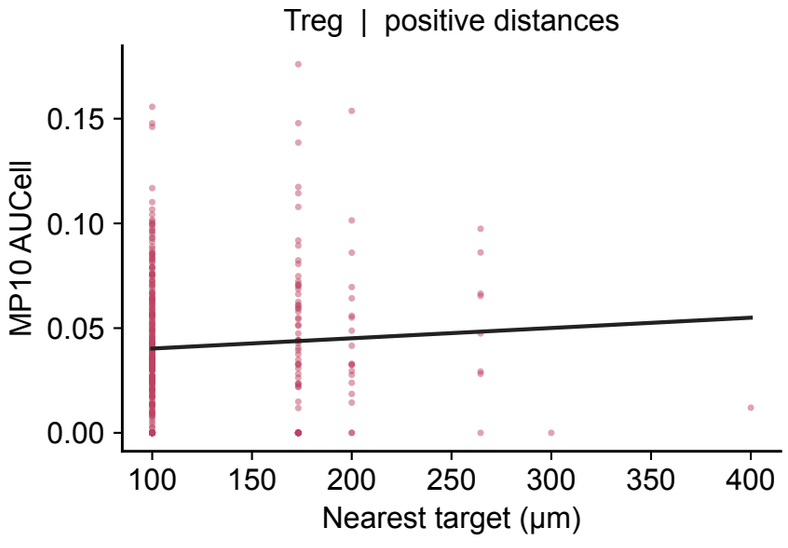
r +0.01 ρ +0.04
n 421/488 queries • 154 targets



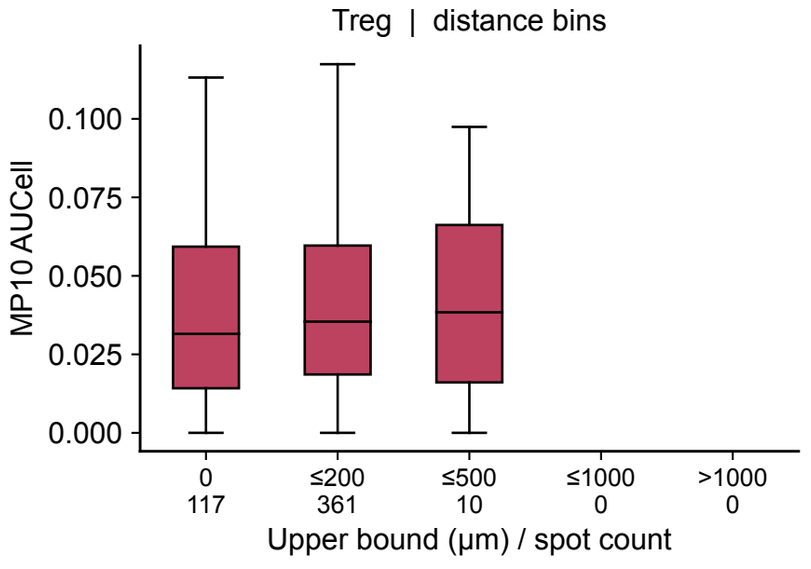
Orange = zero distance
Line = within-section linear fit



r +0.07 ρ +0.06
n 488/488 queries • 154 targets



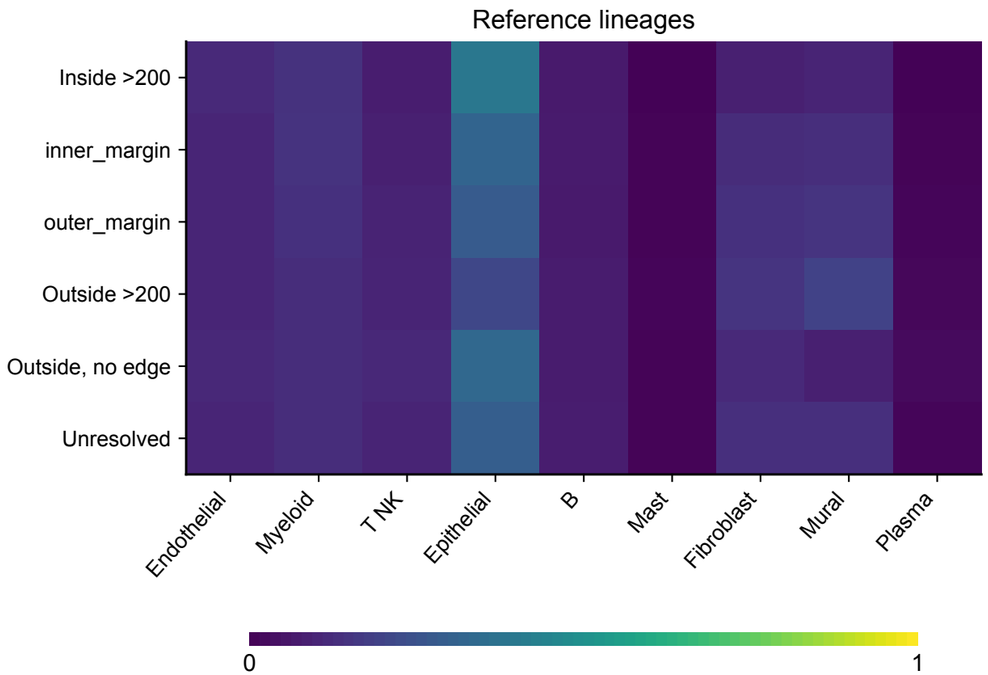
r +0.07 ρ +0.04
n 371/488 queries • 154 targets

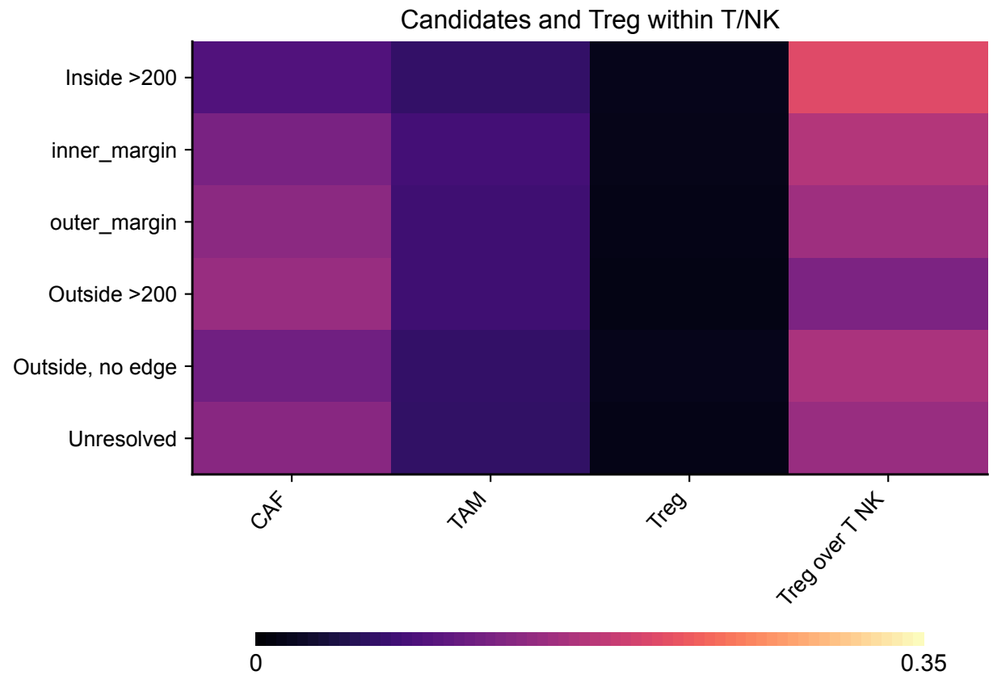


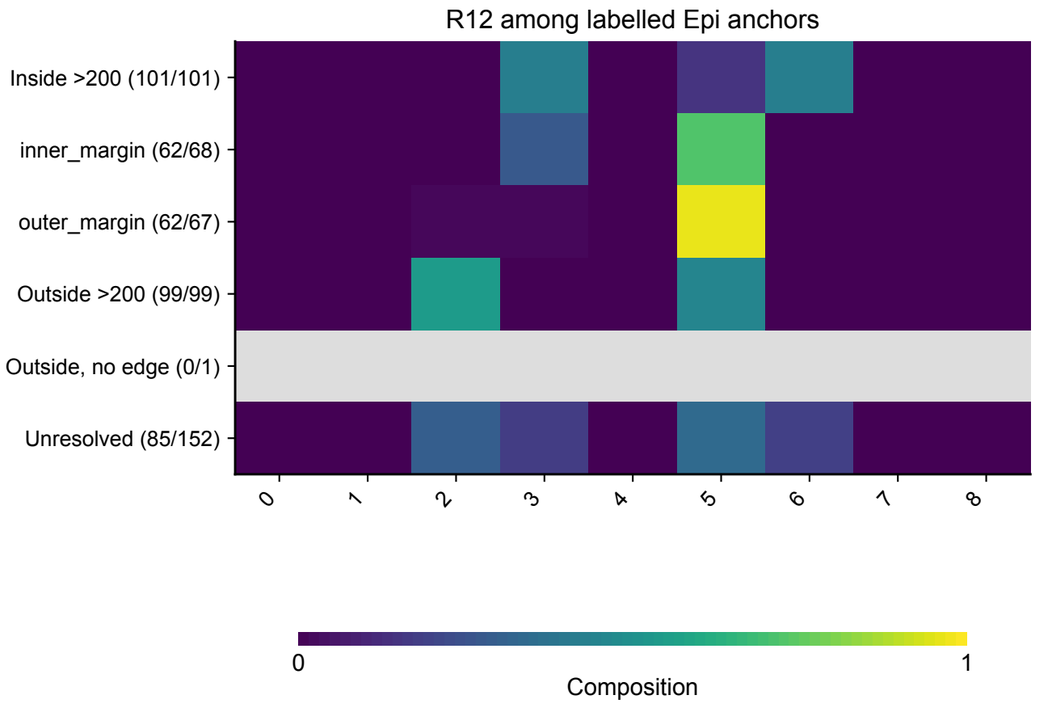
Orange = zero distance
Line = within-section linear fit

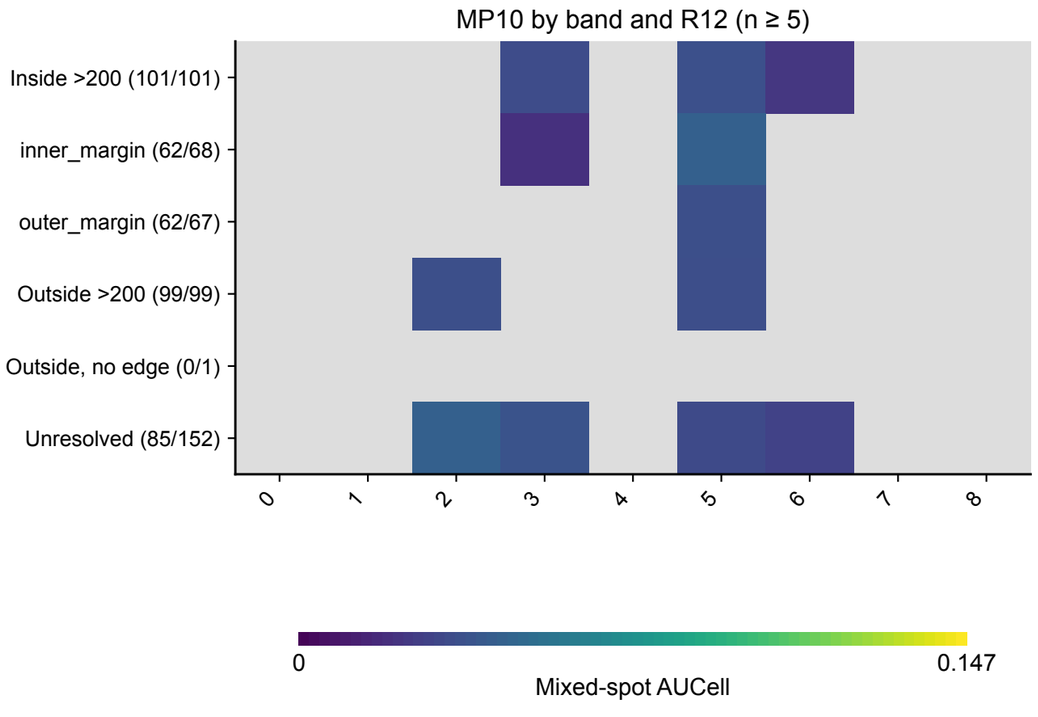
GSE171351_A1 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





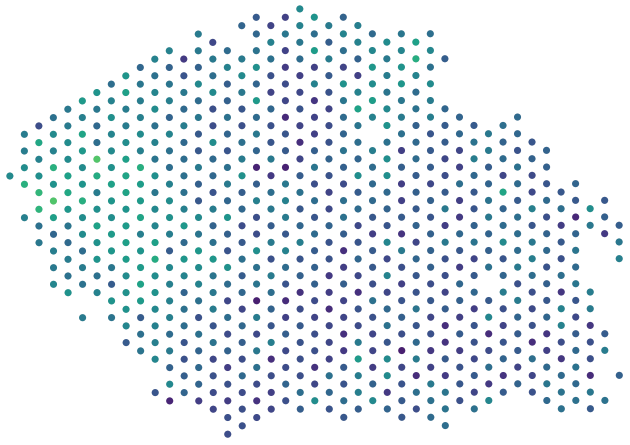




GSE171351_A1 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

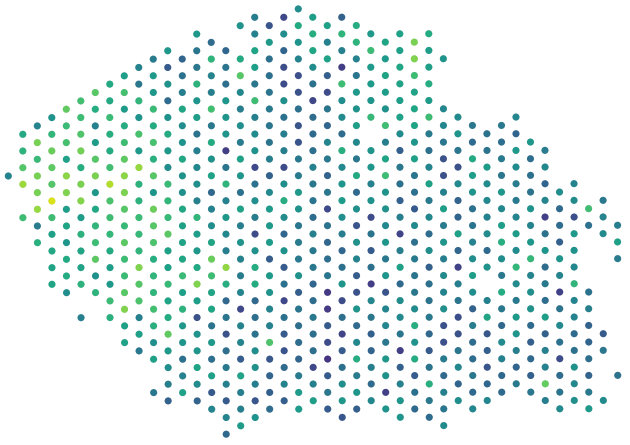
E2F Targets



0 0.152

UCell score

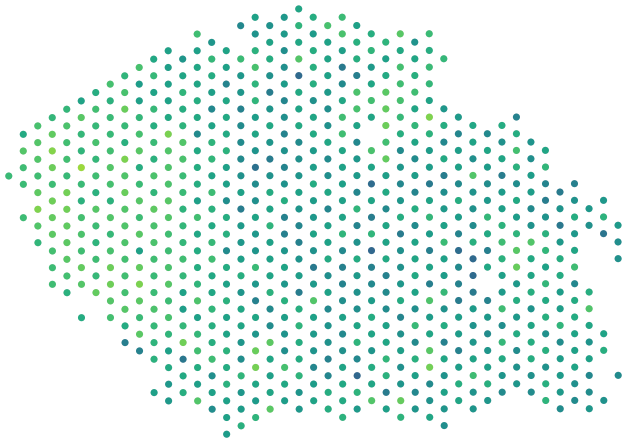
G2M Checkpoint



0 0.134

UCell score

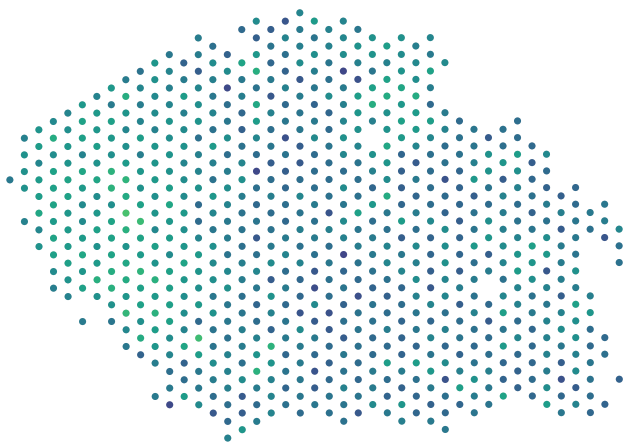
Hypoxia



0 0.211

UCell score

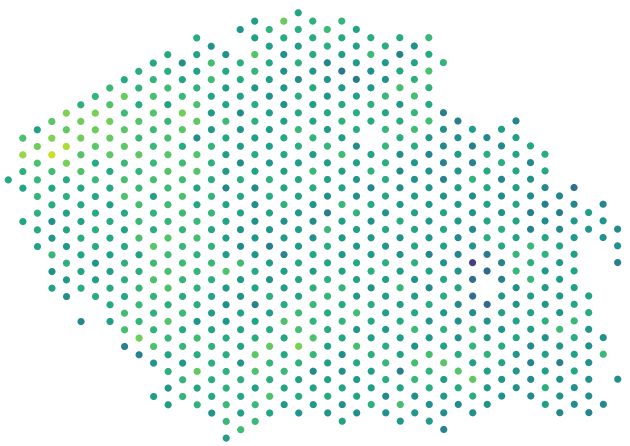
Tnfa Signaling Via Nfkb



0 0.275

UCell score

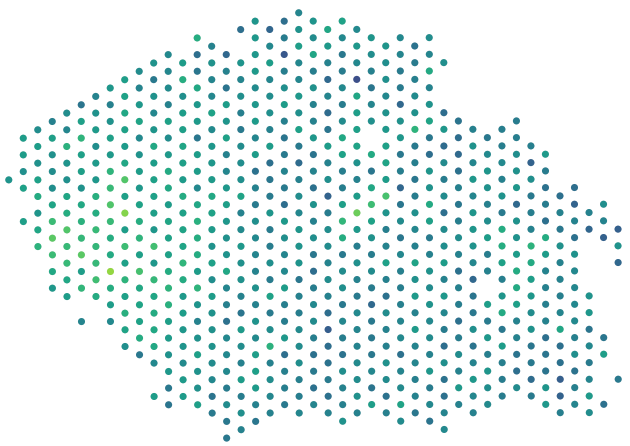
Emt



0 0.386

UCell score

Interferon Gamma Response



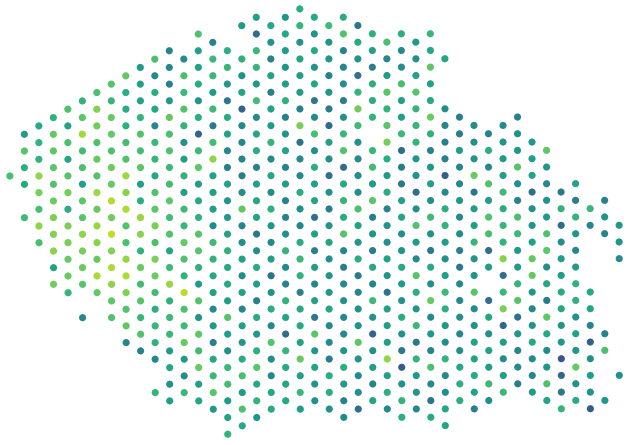
0 0.275

UCell score

GSE171351_A1 | Immune, lipid and matrix programs

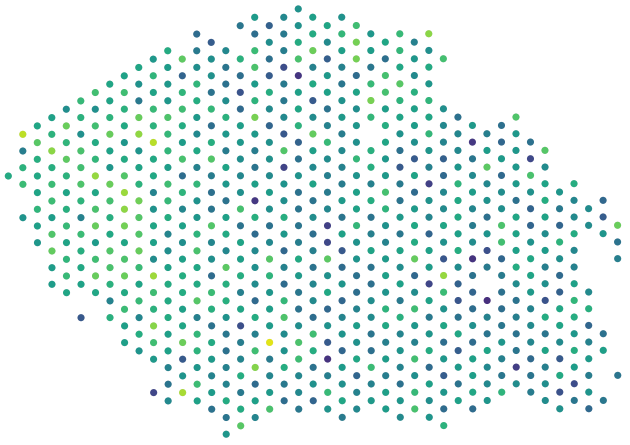
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling



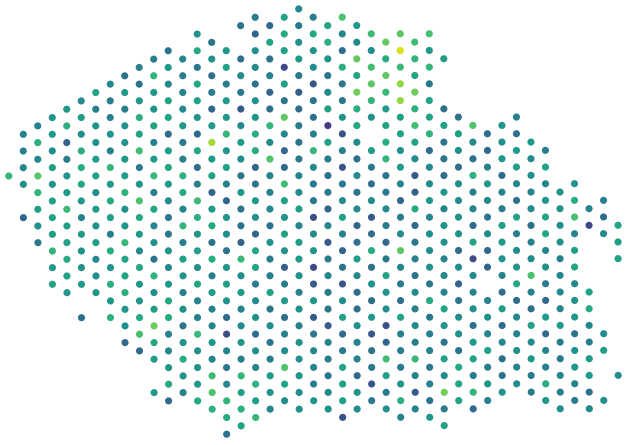


Tgf Beta Signaling



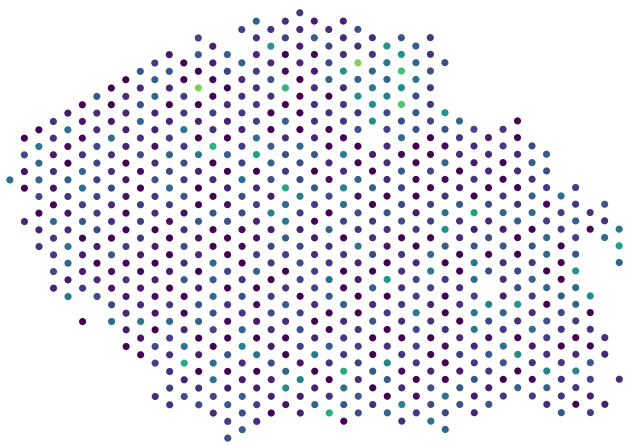


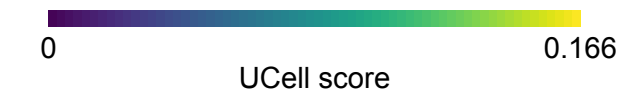
Cholesterol Homeostasis



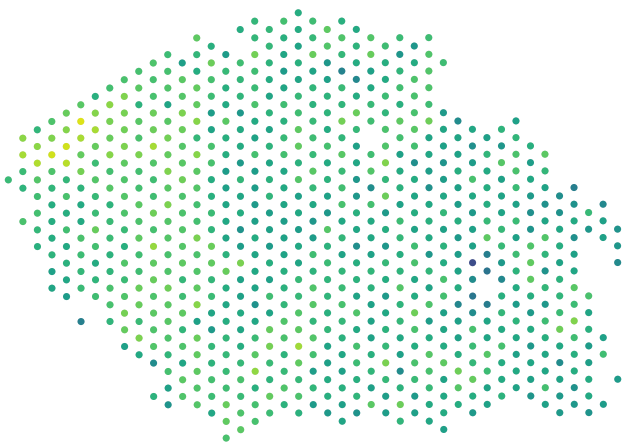


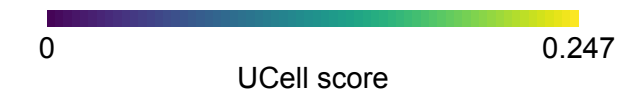
Cholesterol Biosynthesis



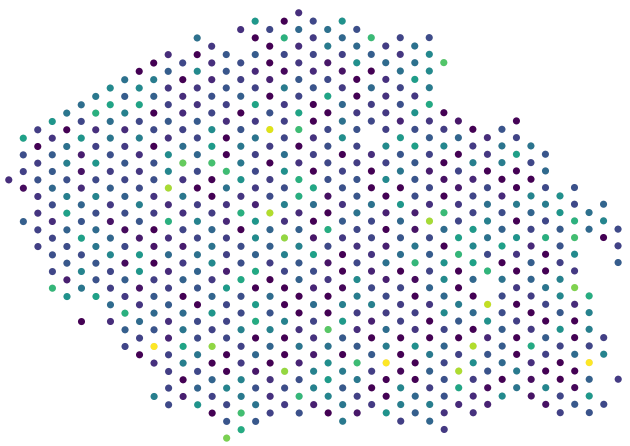


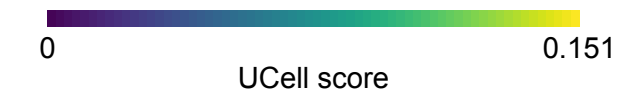
Extracellular Matrix Organization





Co Inhibition By Ctla4

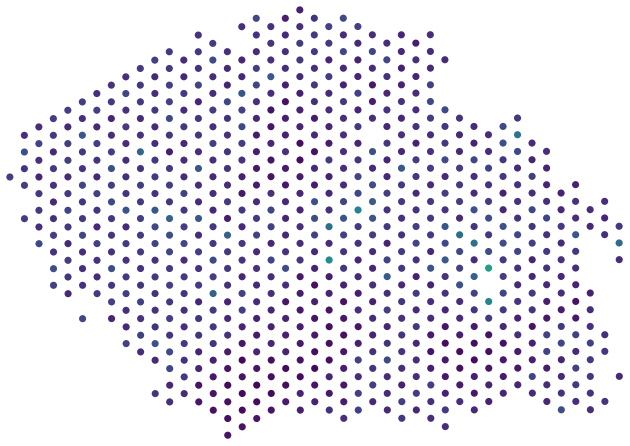




GSE171351_A1 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

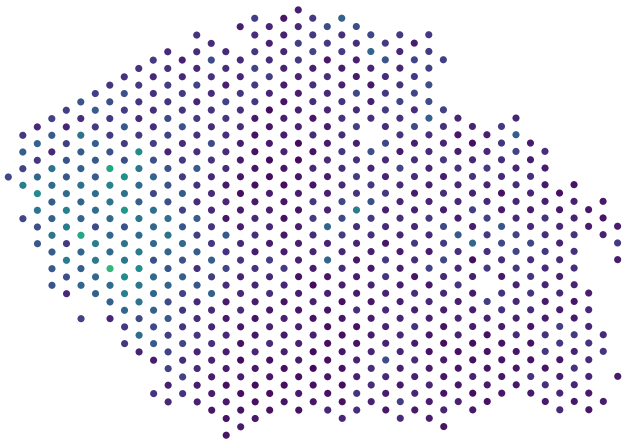
Treg composition



0 0.102

Fraction of all 30

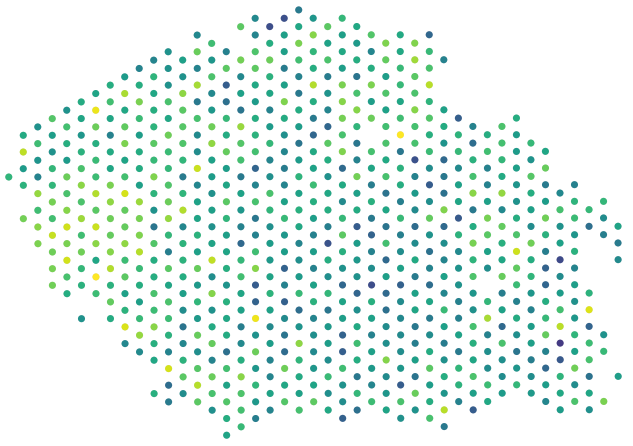
Treg estimated abundance



0 2.85

Posterior mean estimate

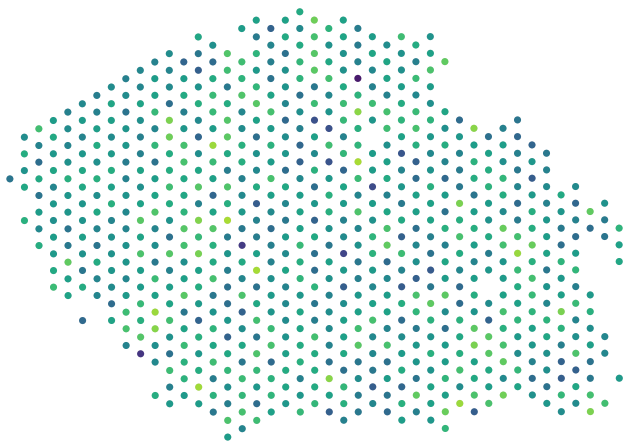
Treg Resting Vs Activated Dn



0 0.0778

UCell score

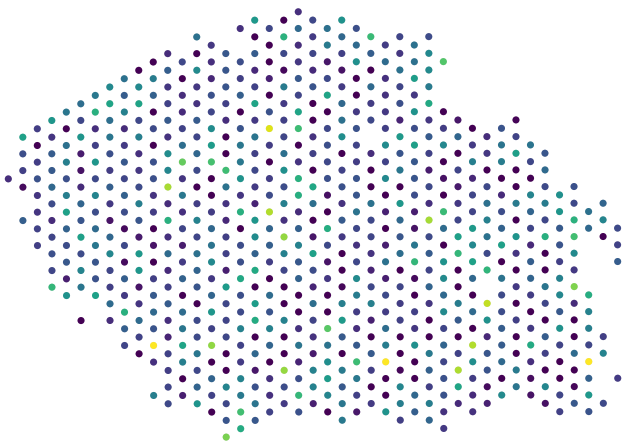
Treg Resting Vs Activated Up



0 0.0609

UCell score

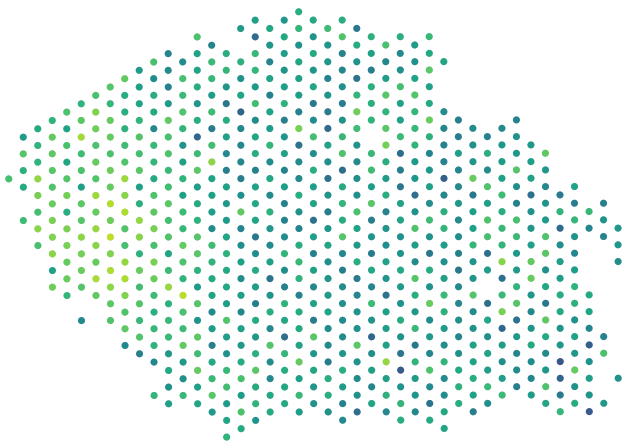
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

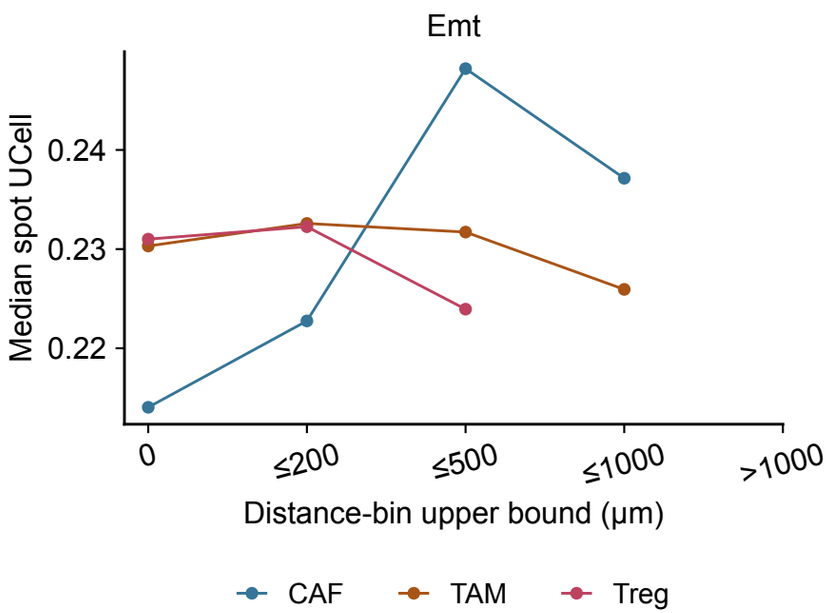


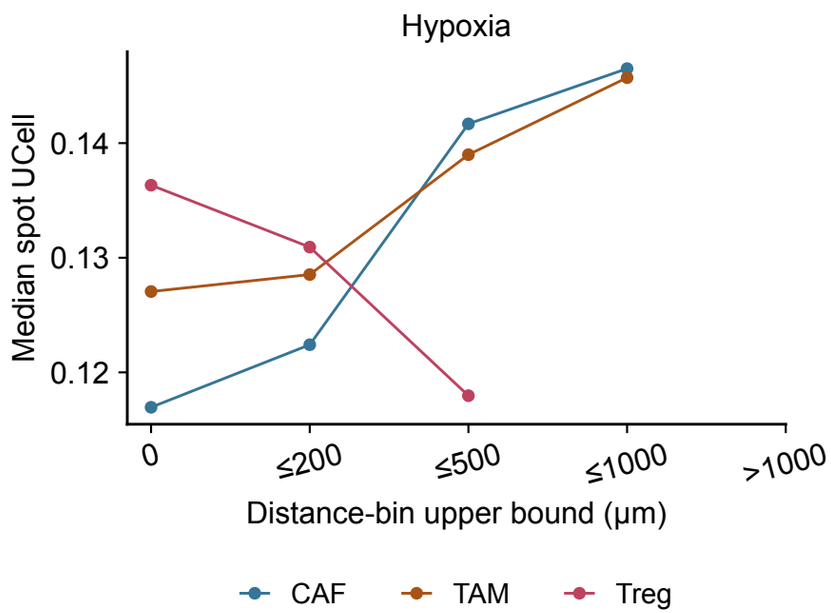
0 0.144

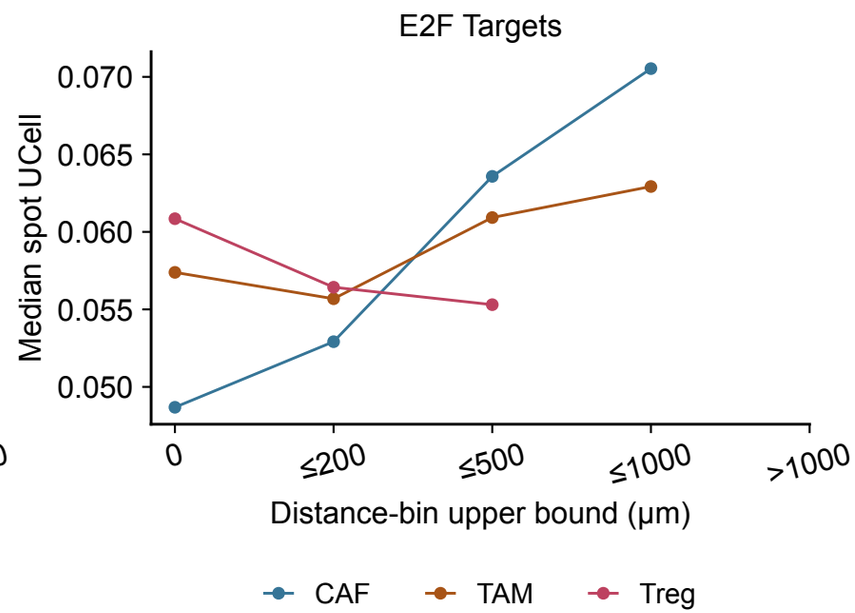
UCell score

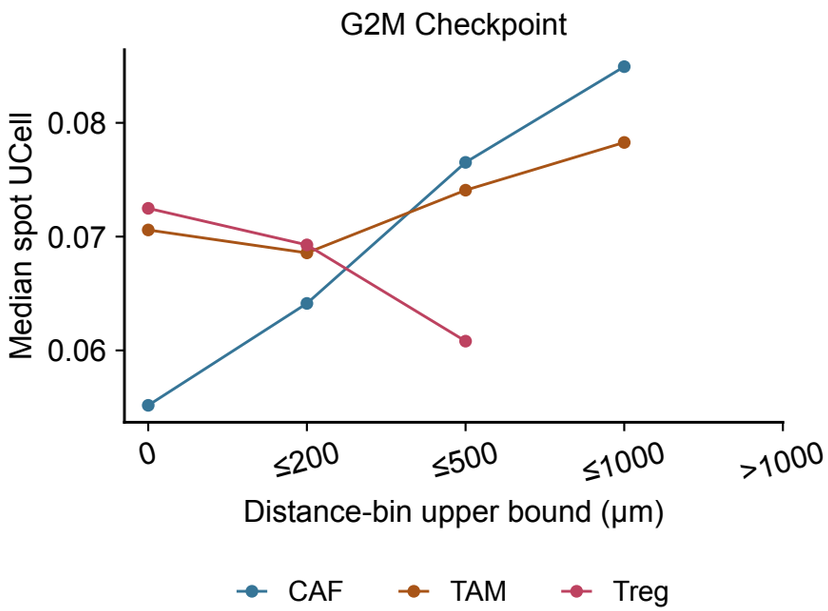
GSE171351_A1 | Programs by target distance

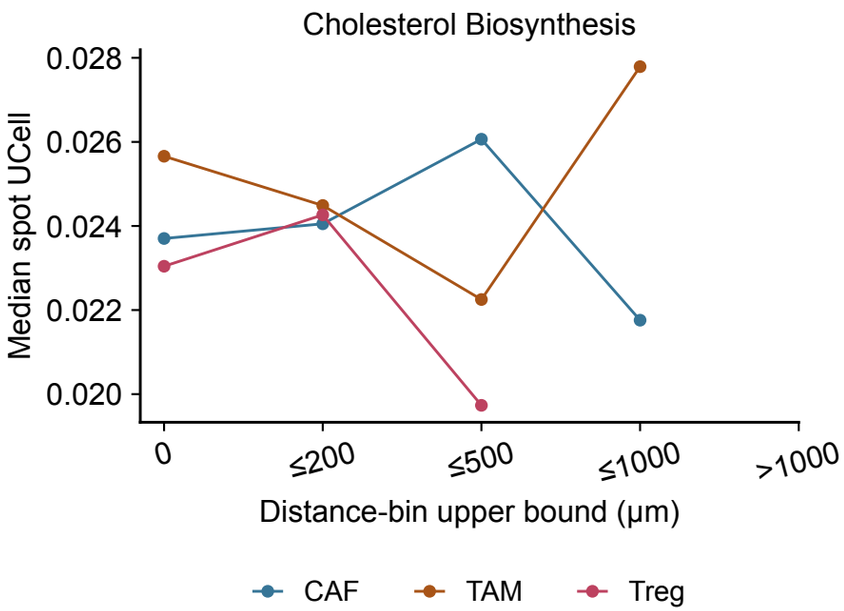
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

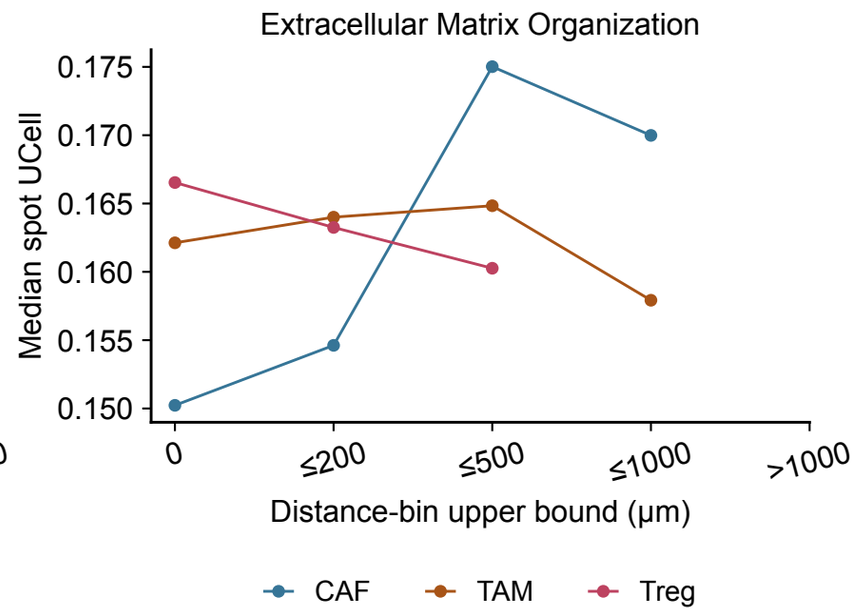






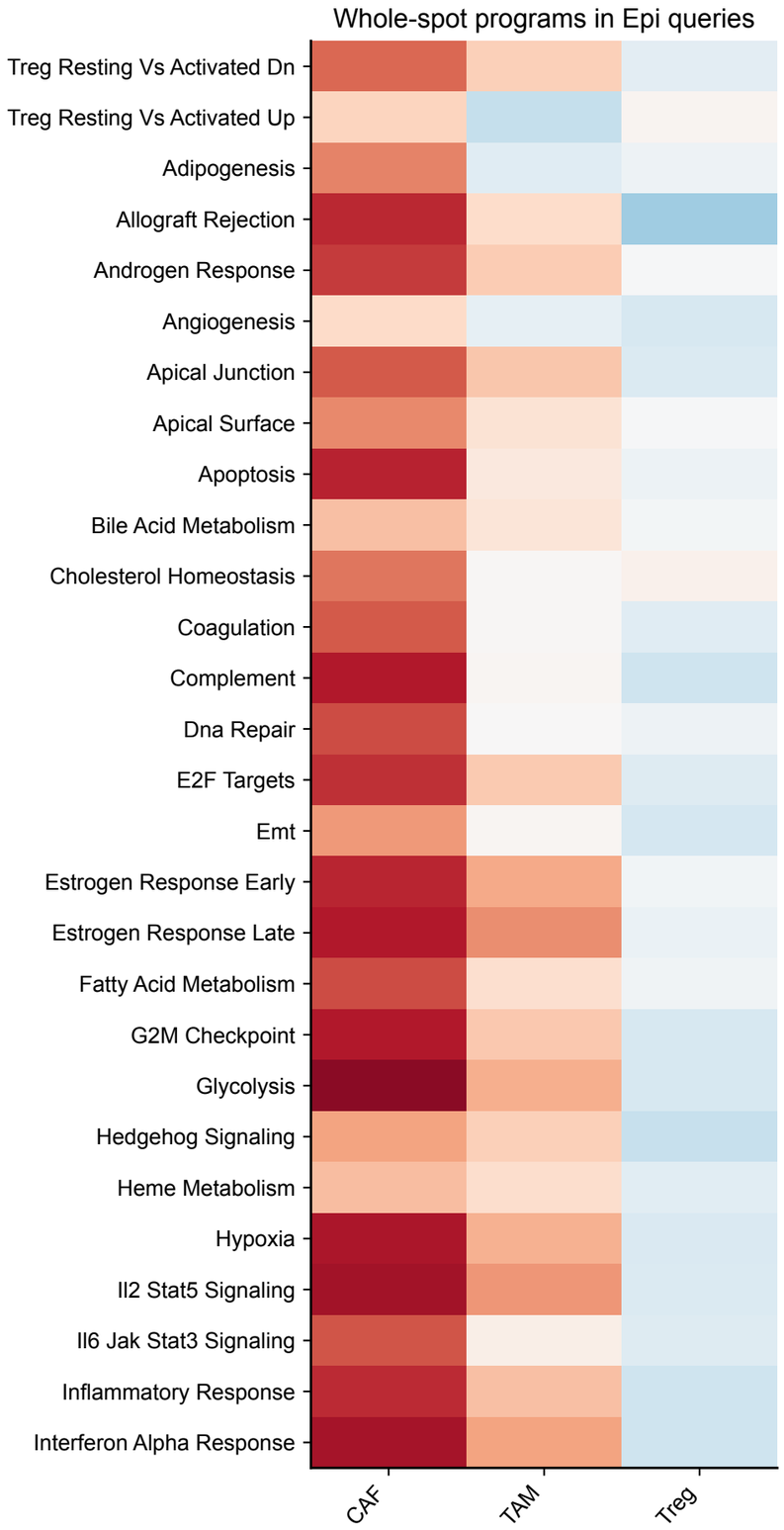


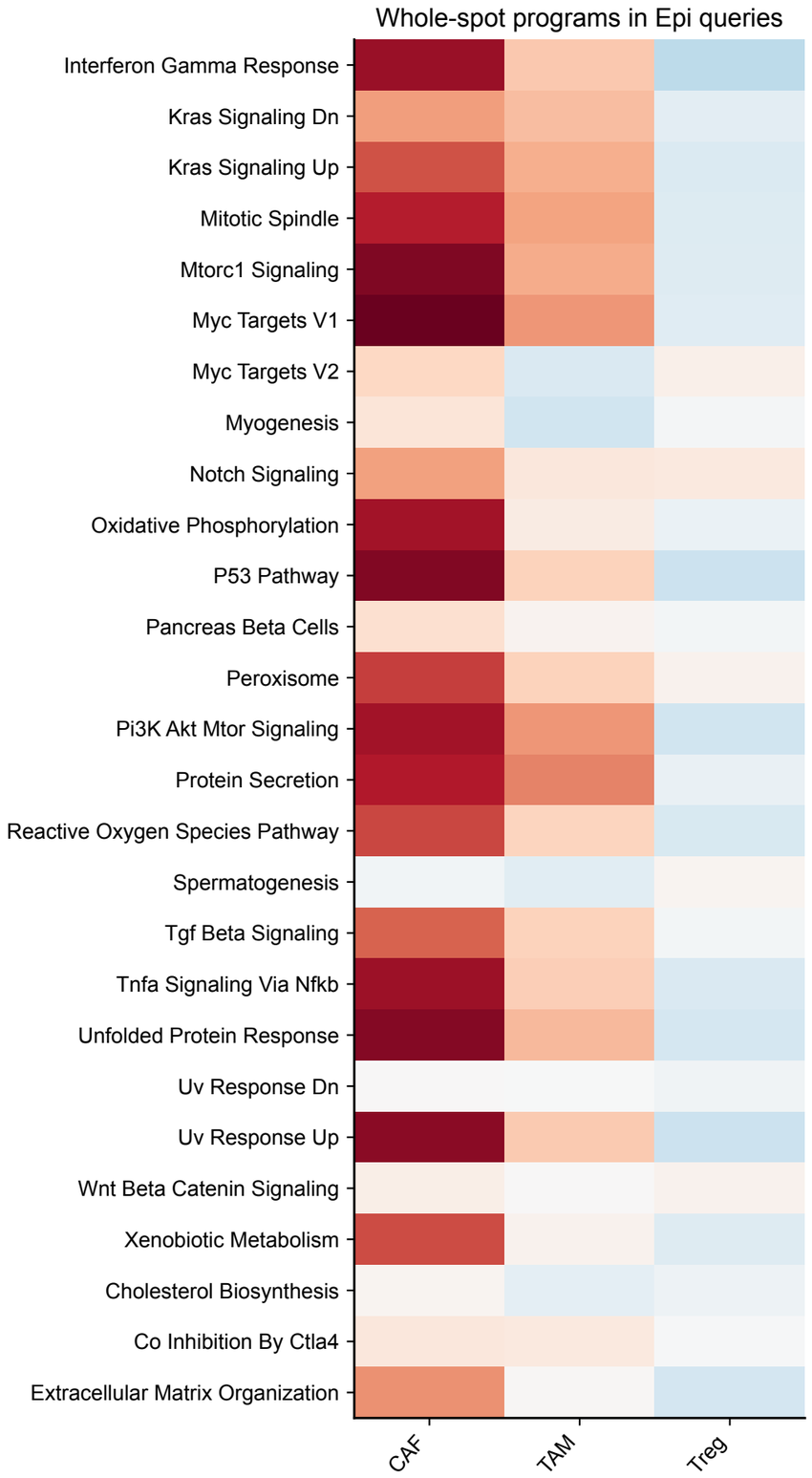




GSE171351_A1 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

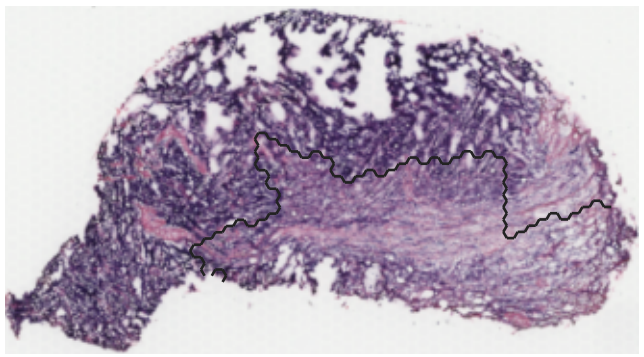




GSE171351_B1 | Tissue context

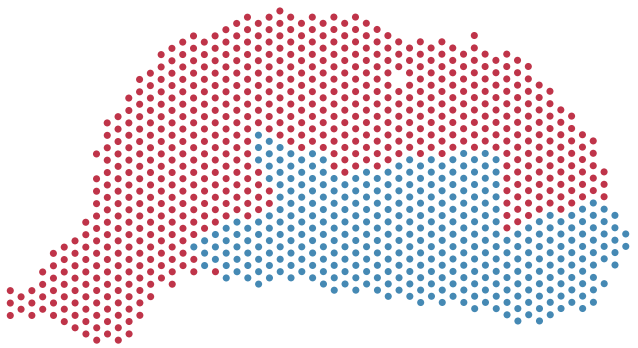
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



Black line: observed candidate outer edge

Final candidate location



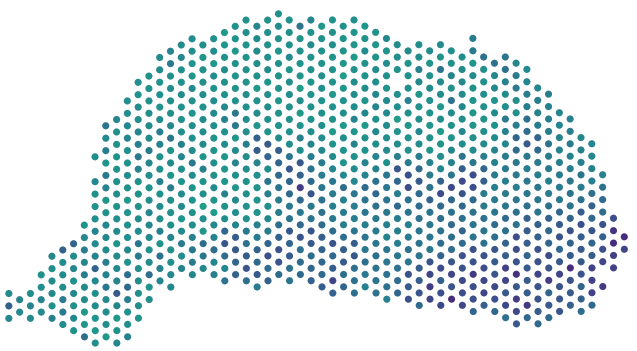
inside outside

Accepted R12 (original 0–8)



1 2 3 4 6

Epithelial support



0 1

Fraction of all 30

MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

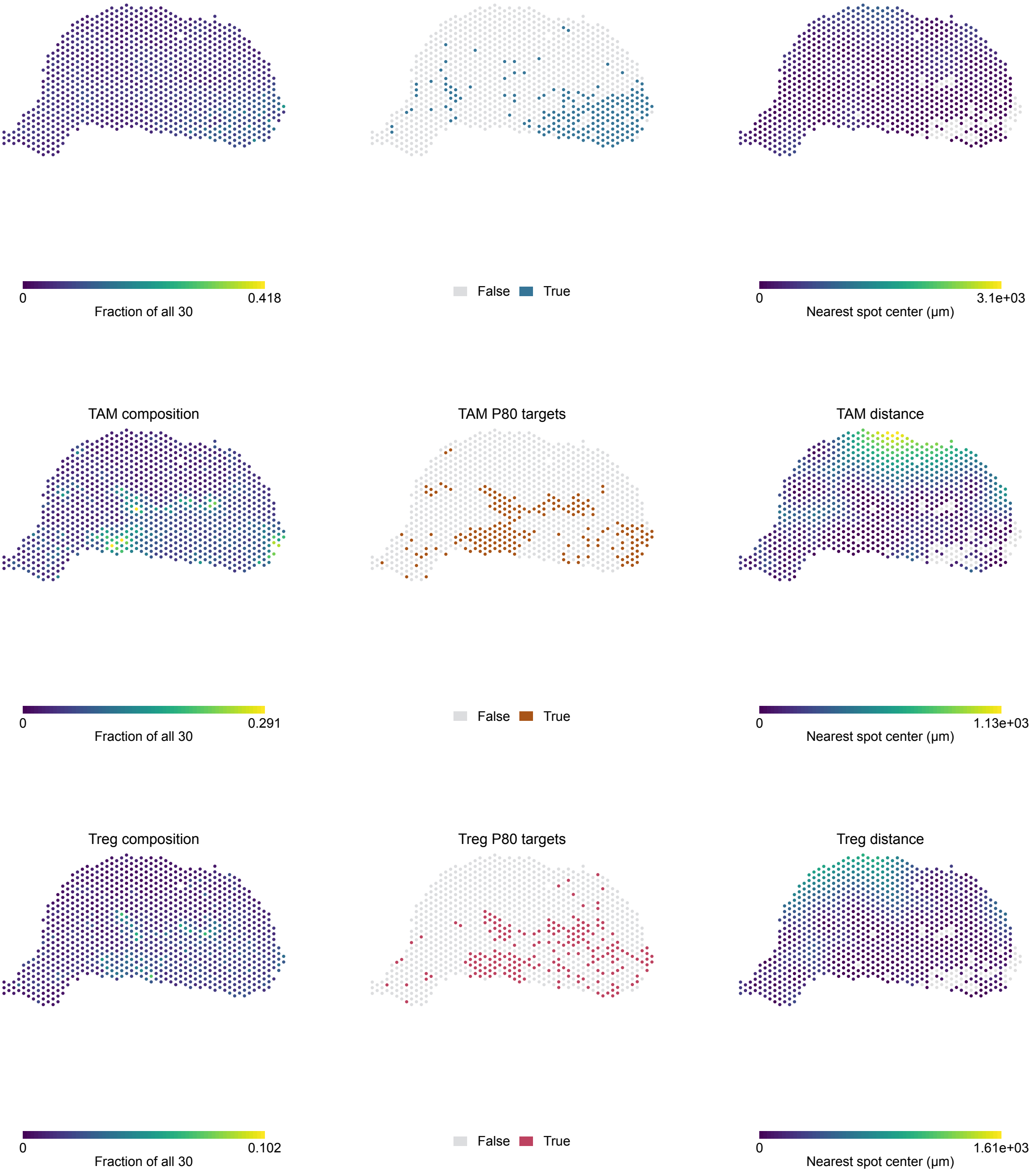


0 0.14

Matched-model AUCell

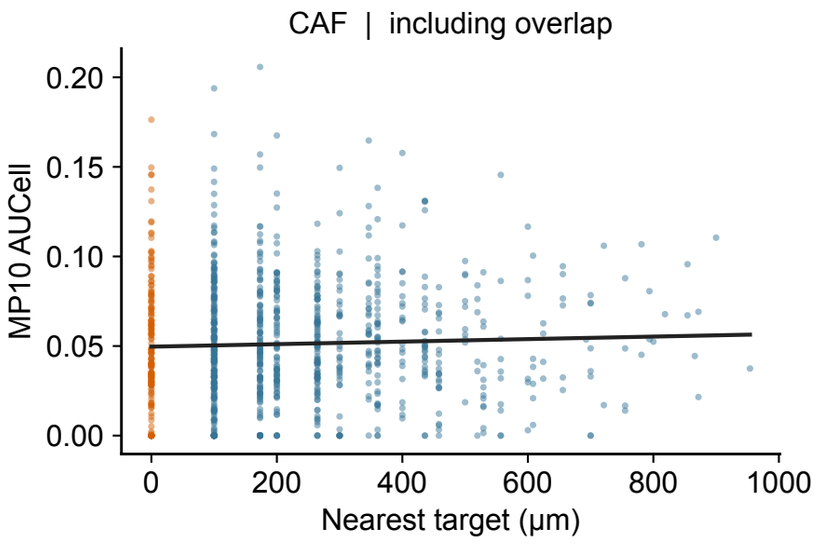
GSE171351_B1 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

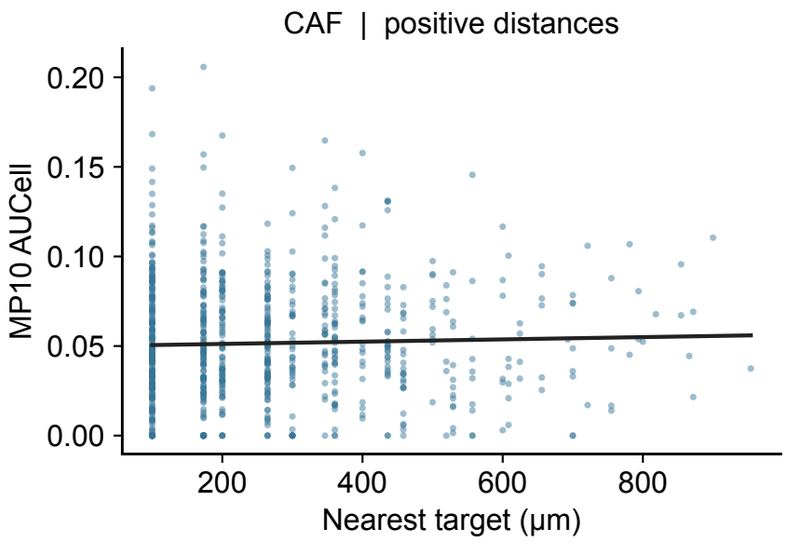


GSE171351_B1 | MP10 versus nearest target

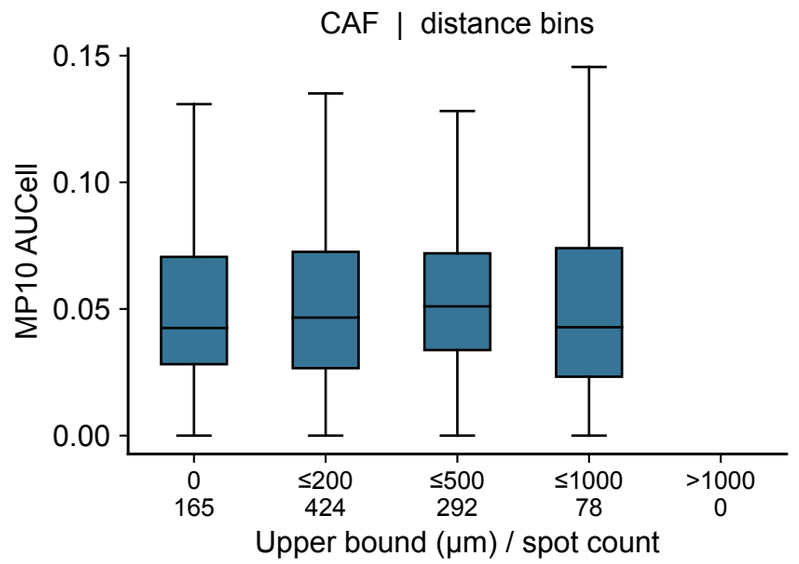
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



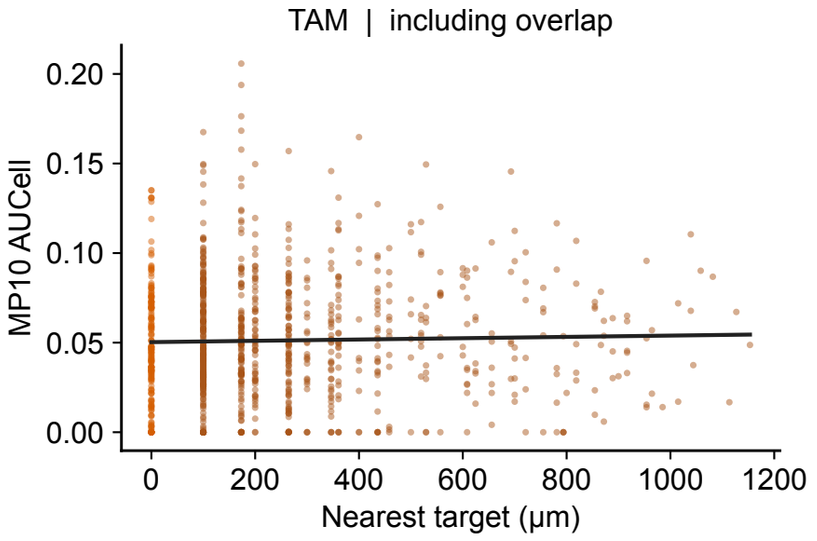
r +0.04 ρ +0.07
n 959/959 queries • 204 targets



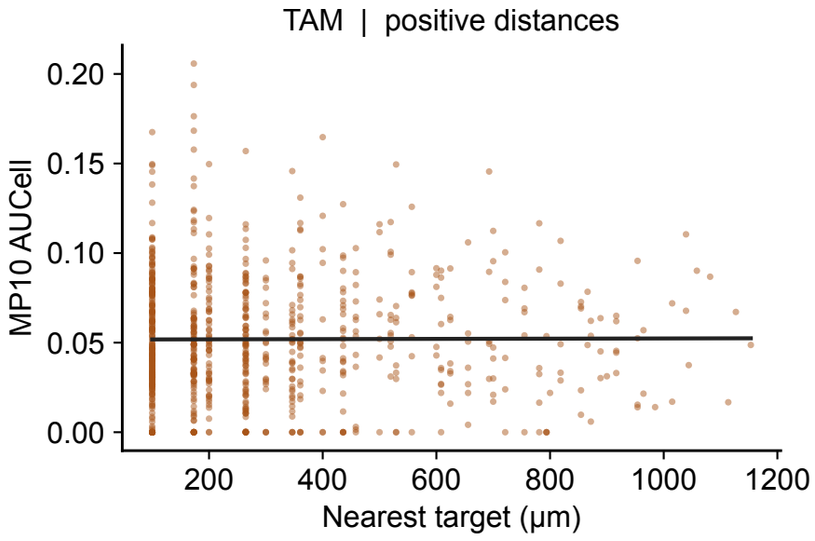
r +0.03 ρ +0.07
n 794/959 queries • 204 targets



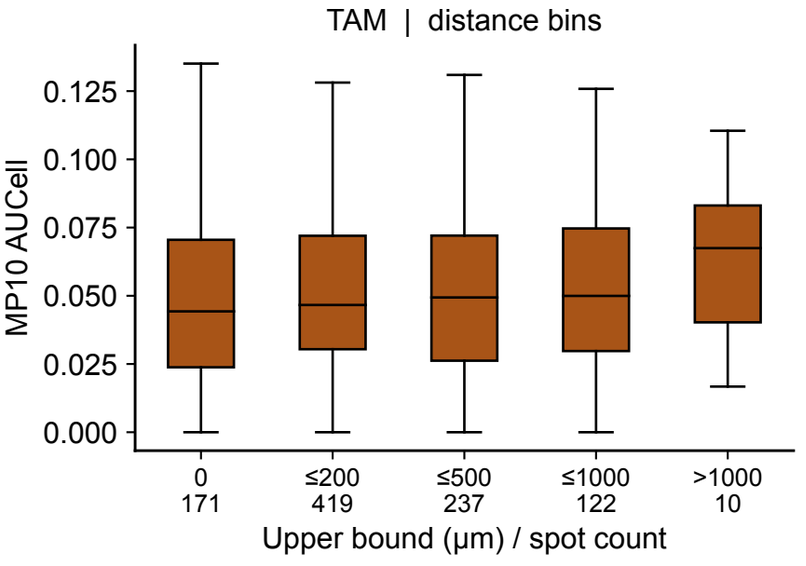
Orange = zero distance
Line = within-section linear fit



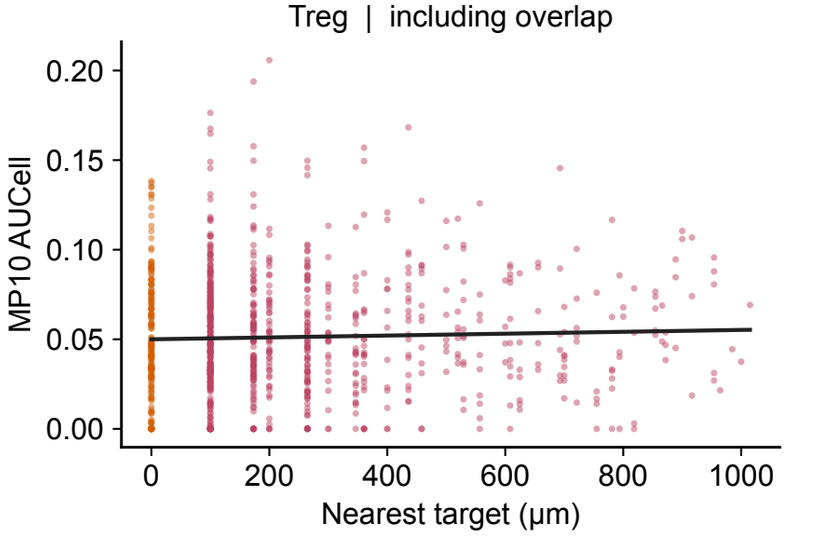
r +0.03 ρ +0.05
n 959/959 queries • 204 targets



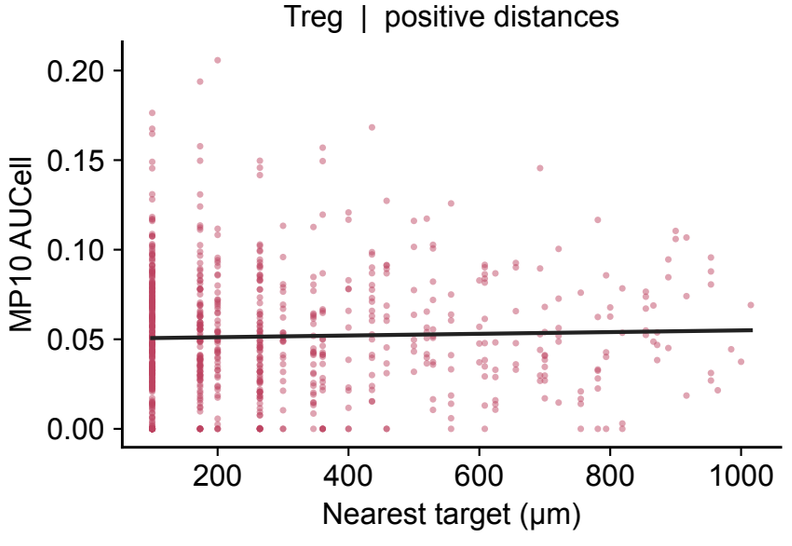
r +0.00 ρ +0.03
n 788/959 queries • 204 targets



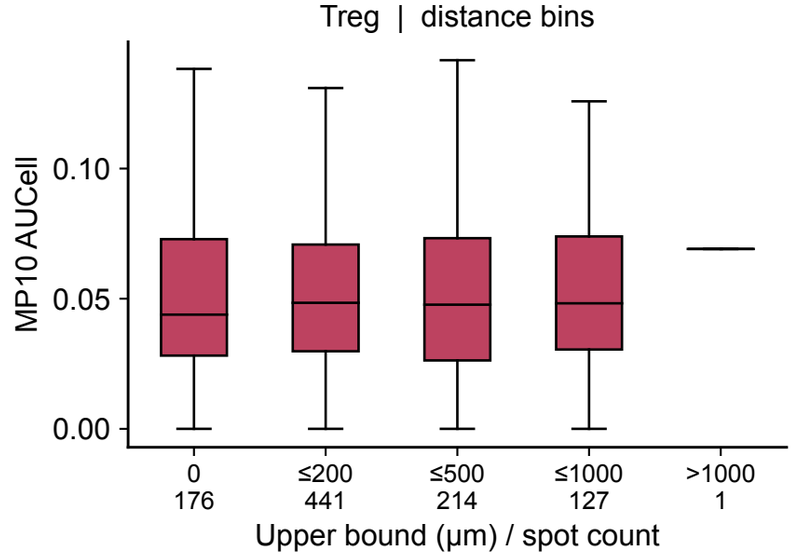
Orange = zero distance
Line = within-section linear fit



r +0.04 ρ +0.03
n 959/959 queries • 204 targets



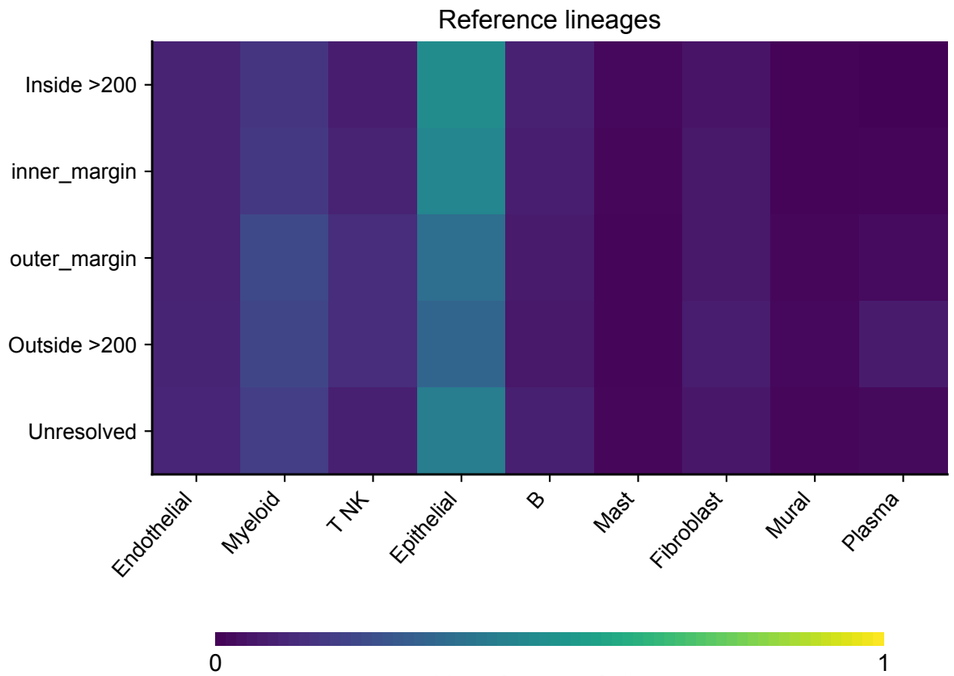
r +0.03 ρ +0.03
n 783/959 queries • 204 targets

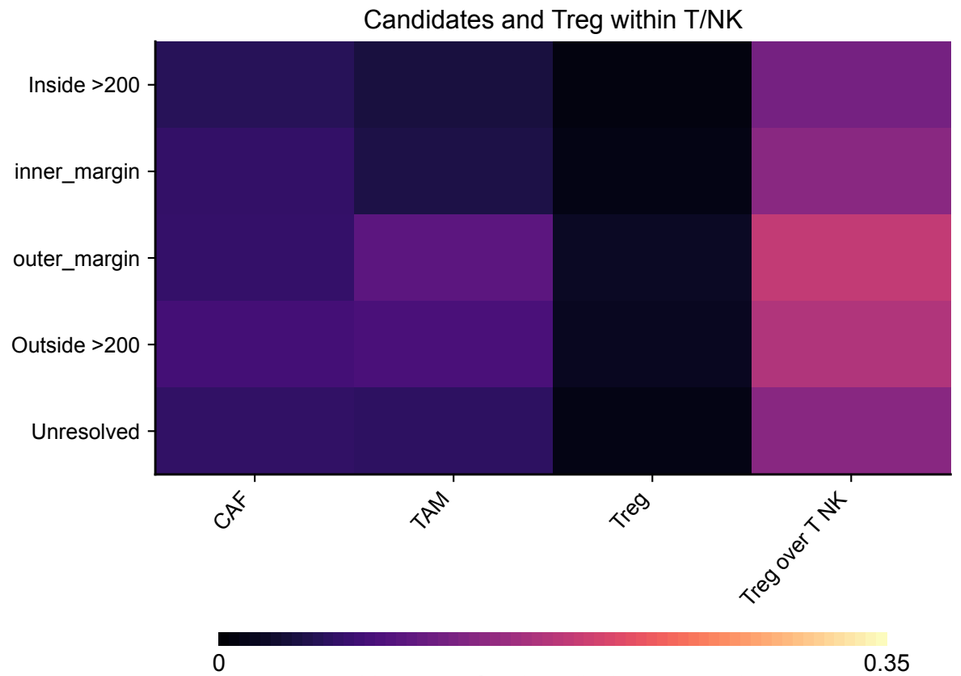


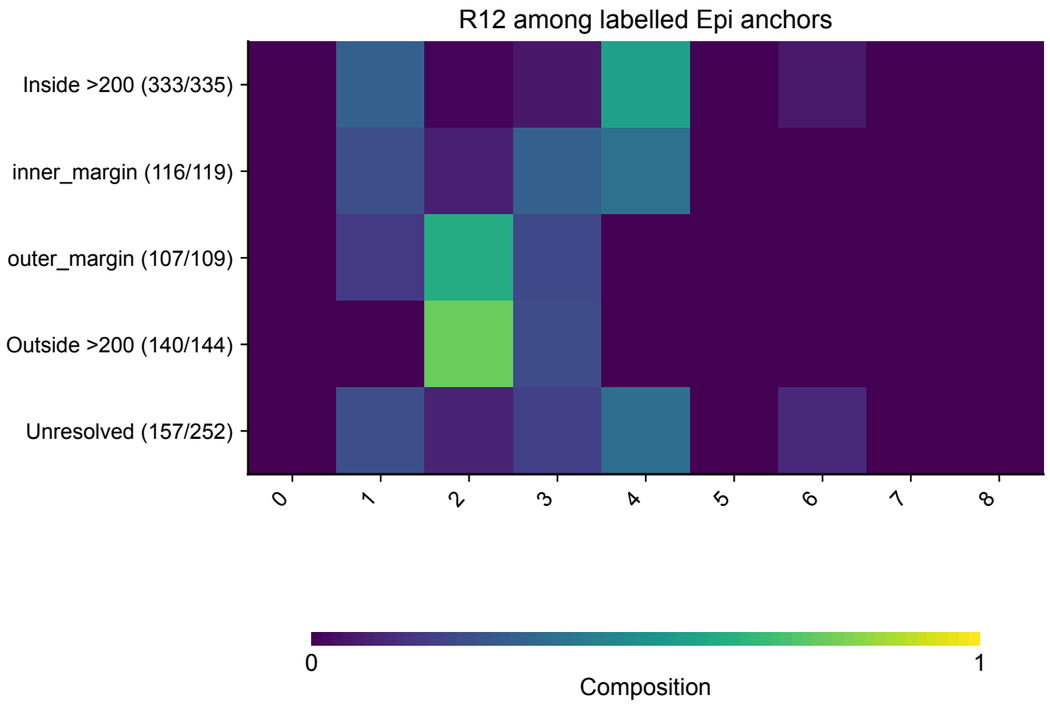
Orange = zero distance
Line = within-section linear fit

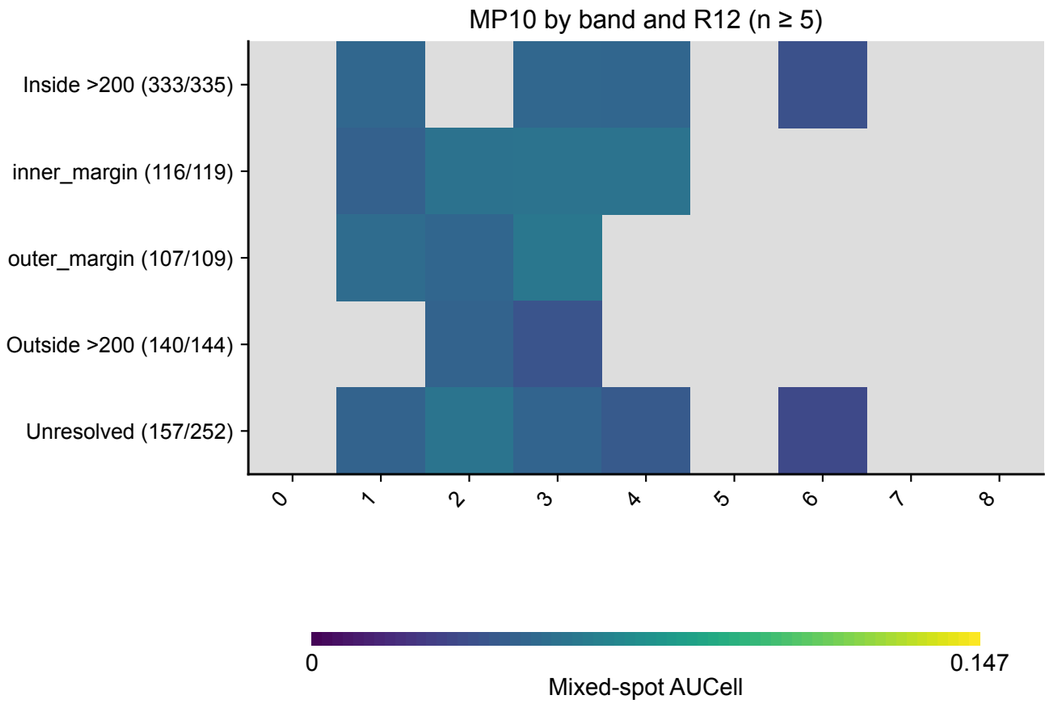
GSE171351_B1 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable



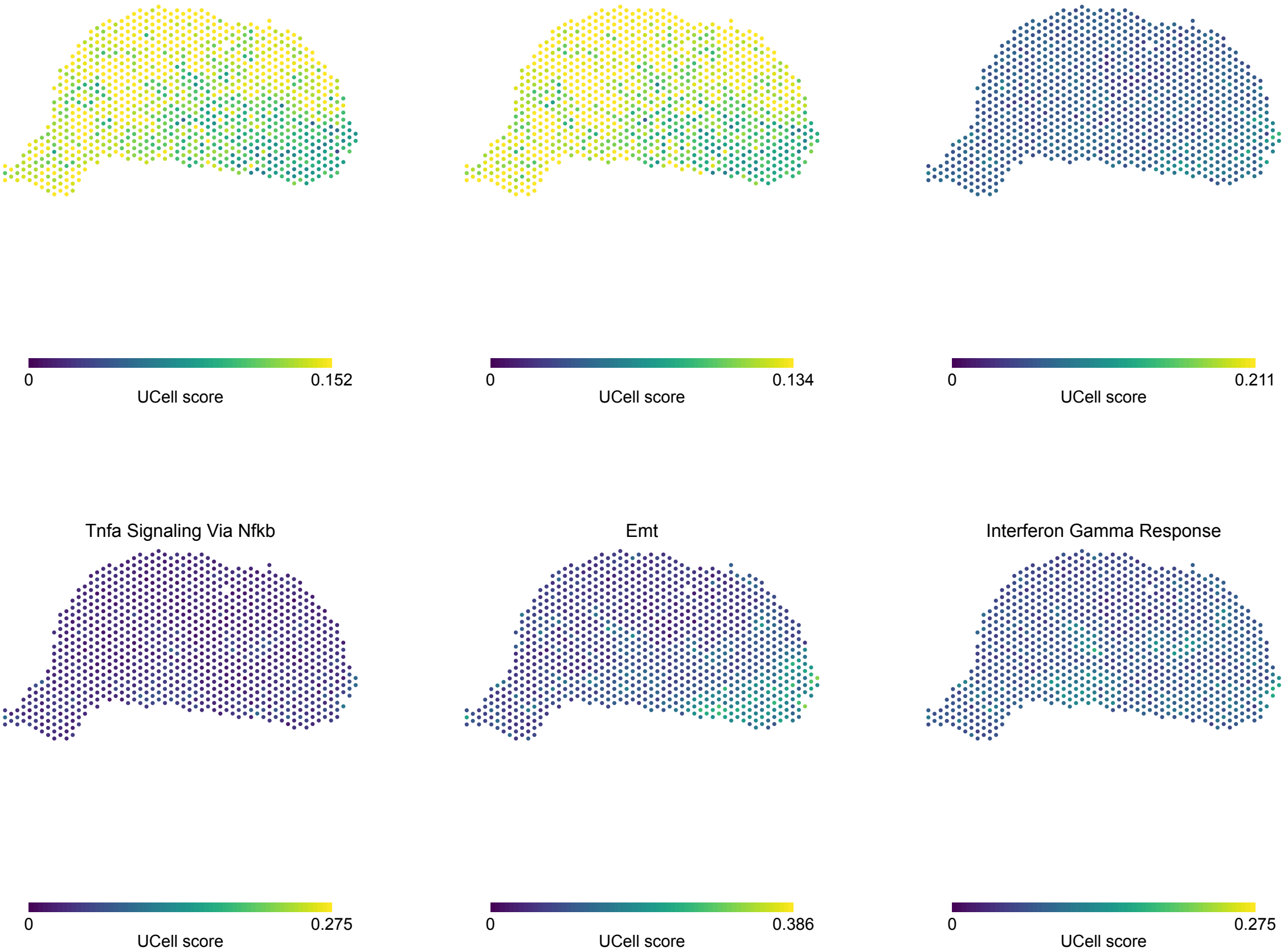






GSE171351_B1 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections



GSE171351_B1 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling



0 0.144

UCell score

Tgf Beta Signaling



0 0.232

UCell score

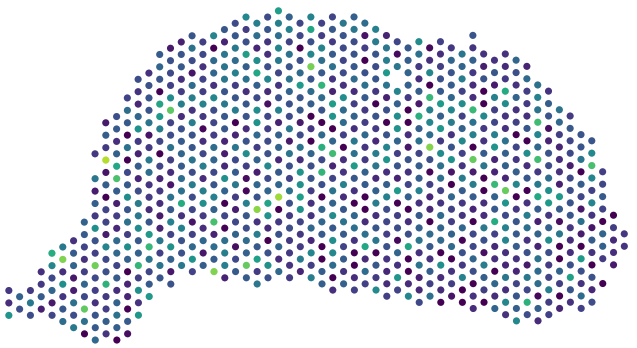
Cholesterol Homeostasis



0 0.21

UCell score

Cholesterol Biosynthesis



0 0.166

UCell score

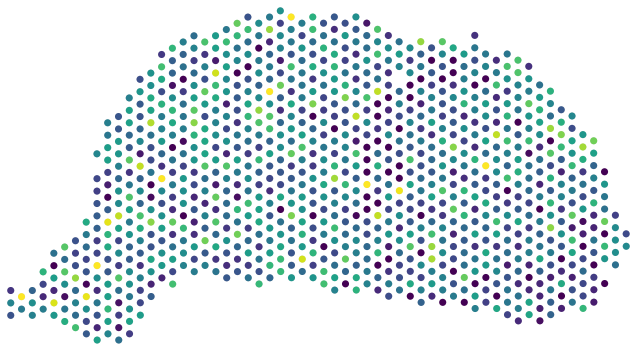
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



0 0.151

UCell score

GSE171351_B1 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition



0 0.102

Fraction of all 30

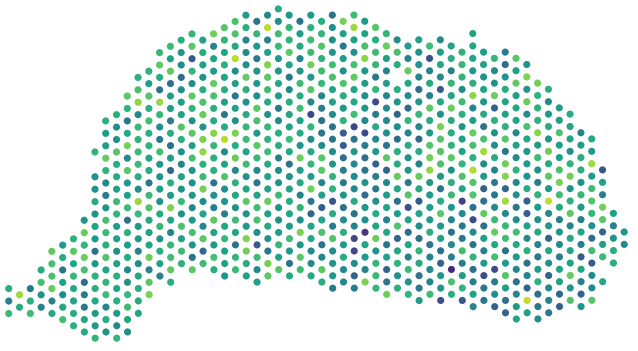
Treg estimated abundance



0 2.85

Posterior mean estimate

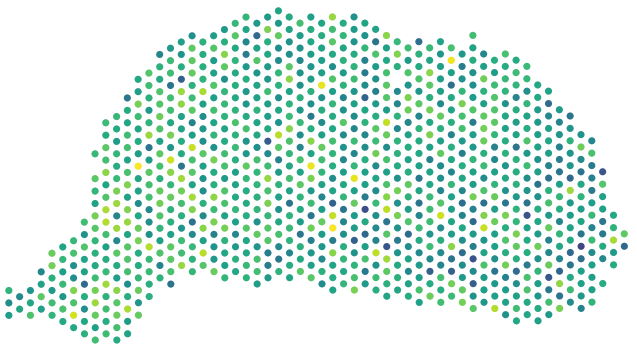
Treg Resting Vs Activated Dn



0 0.0778

UCell score

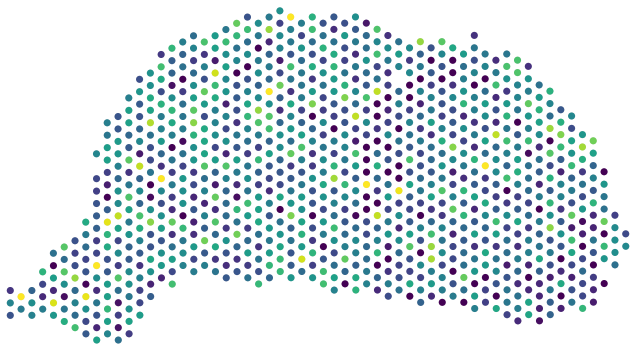
Treg Resting Vs Activated Up



0 0.0609

UCell score

Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

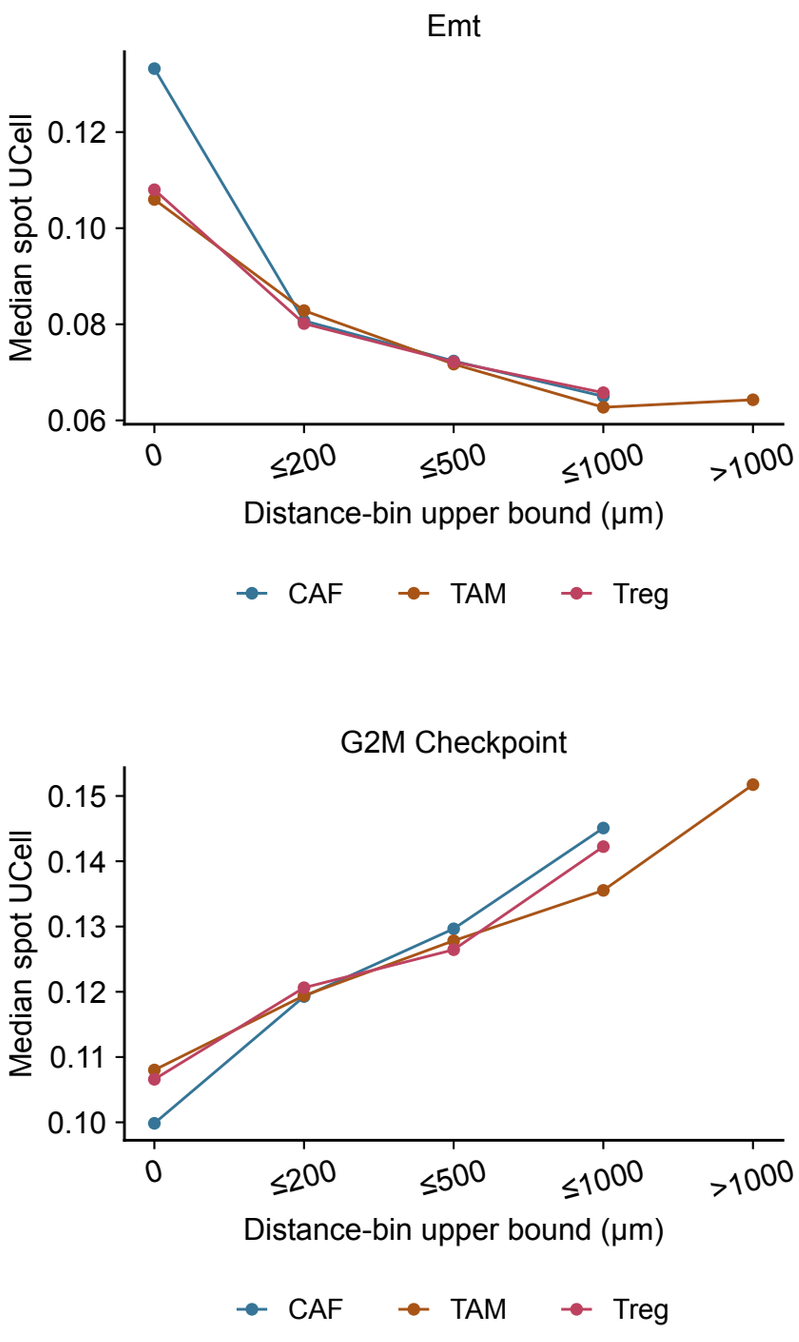


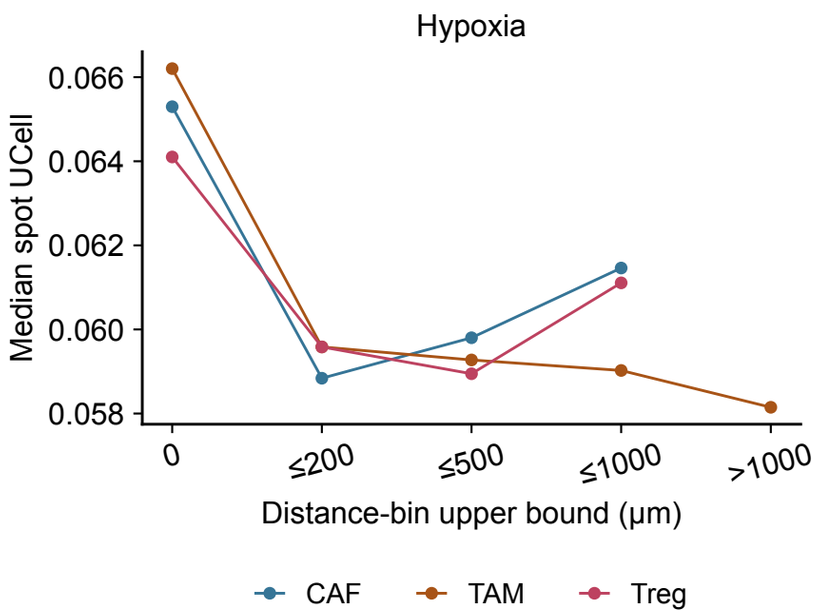
0 0.144

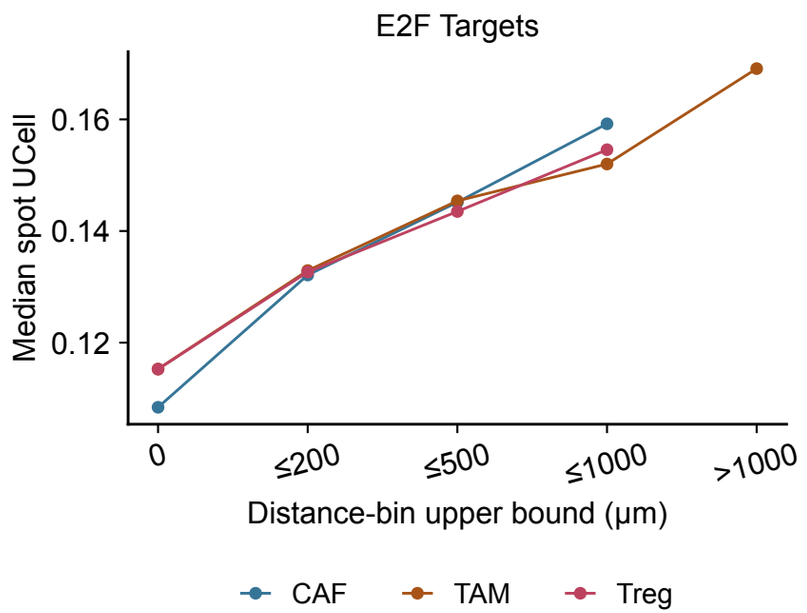
UCell score

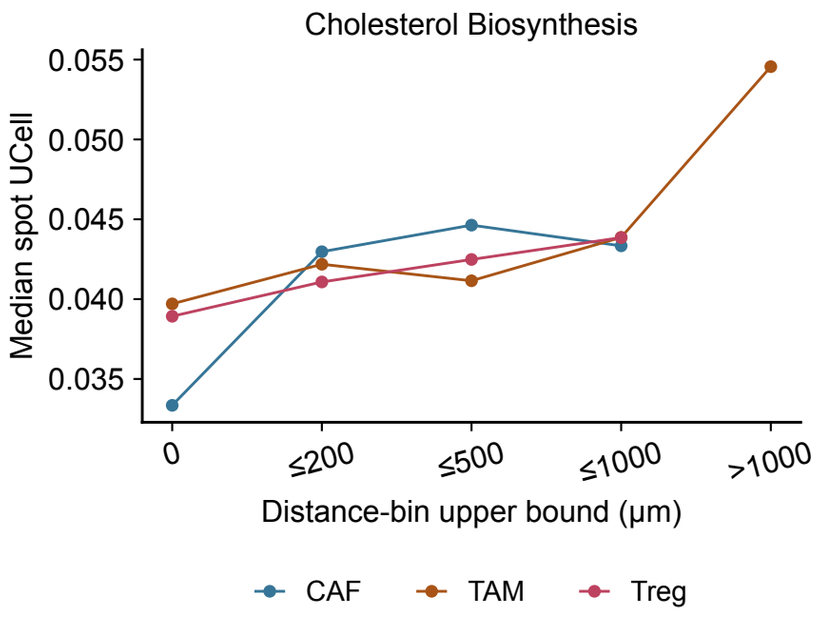
GSE171351_B1 | Programs by target distance

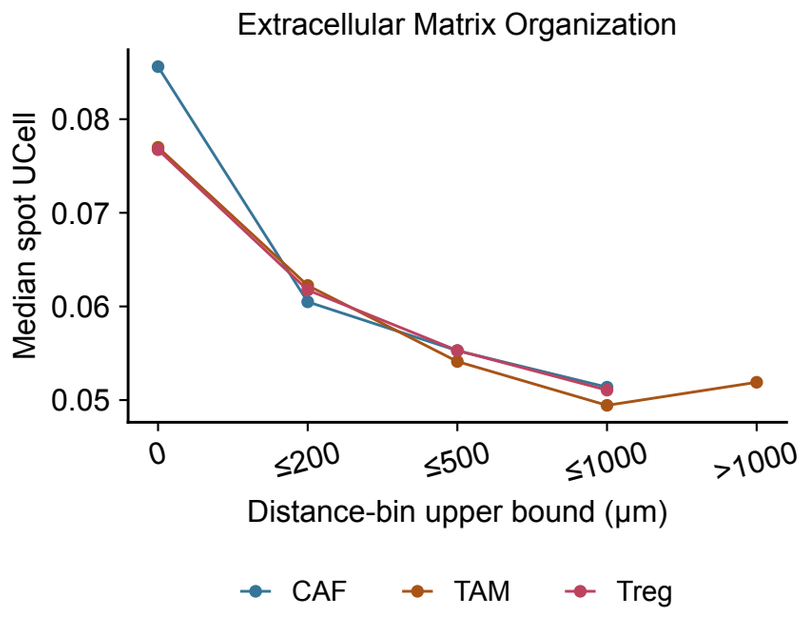
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots





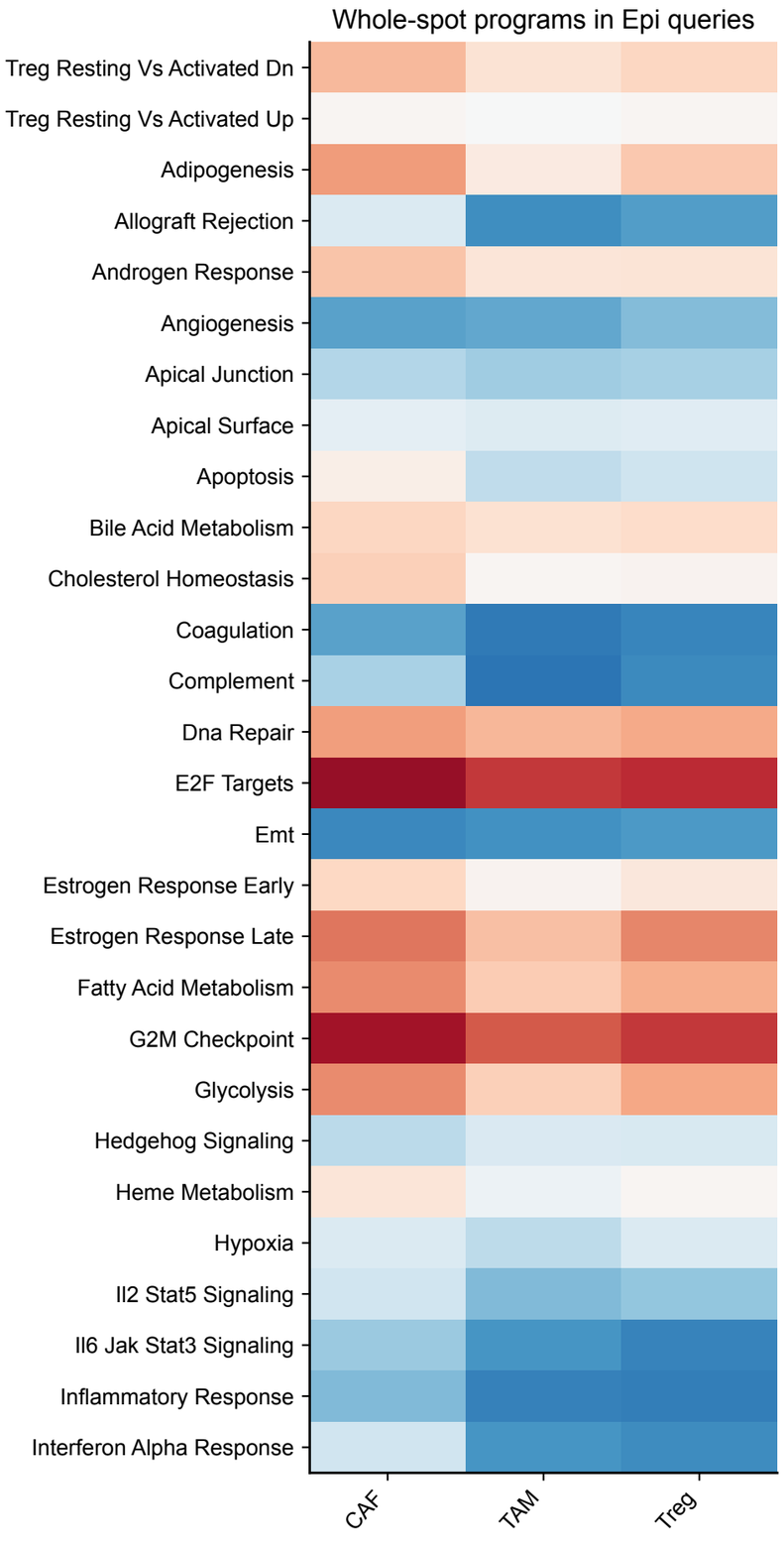


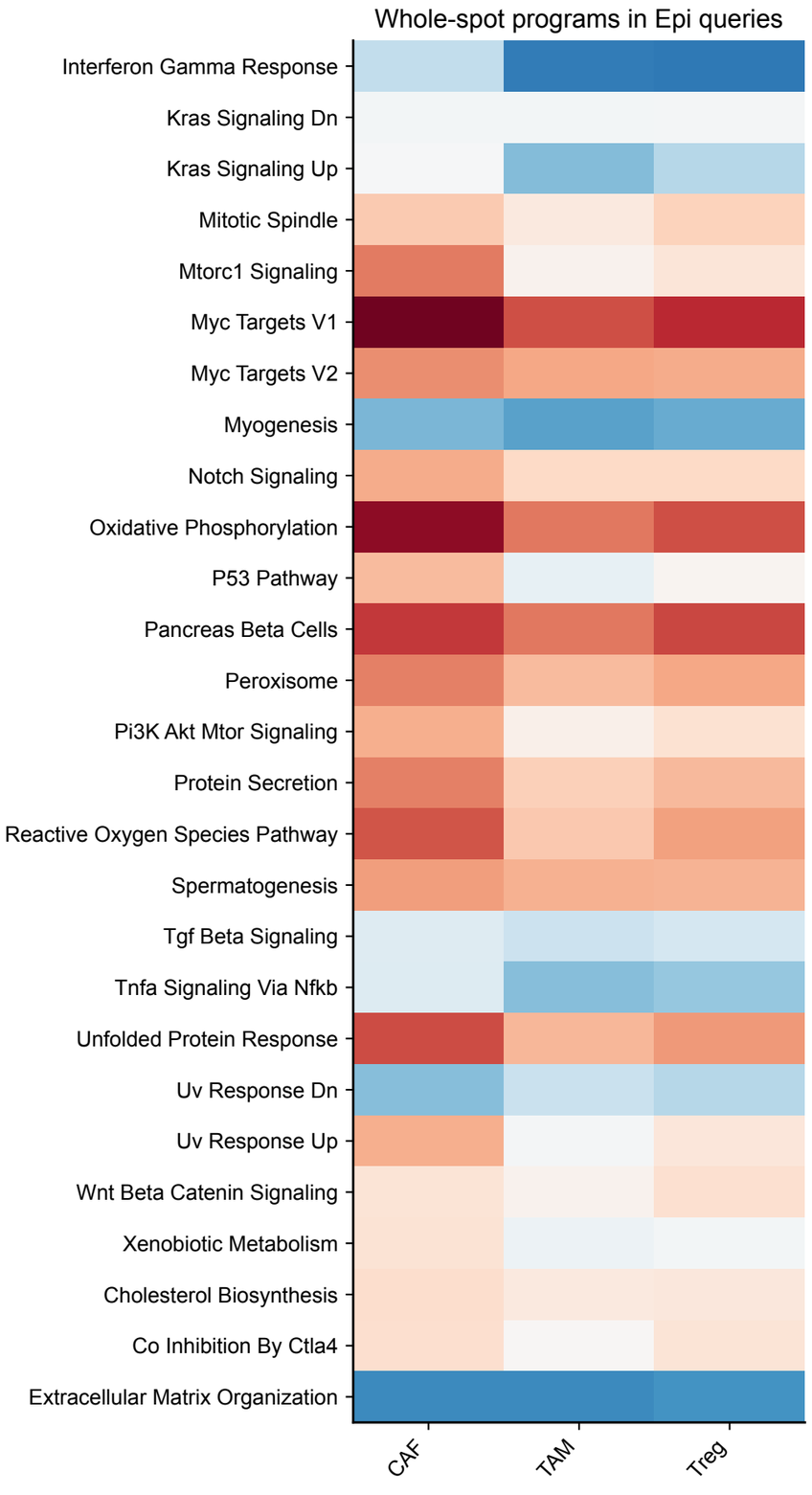




GSE171351_B1 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6





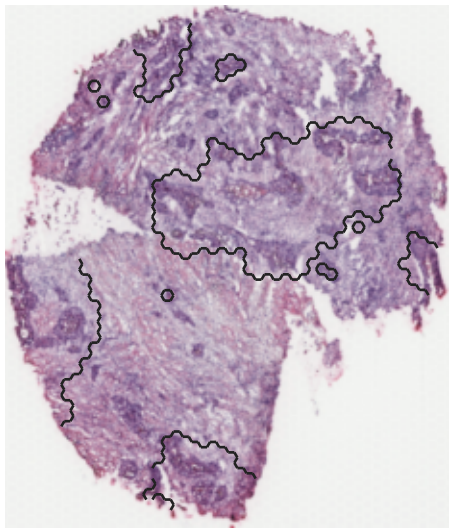




GSE171351_C1 | Tissue context

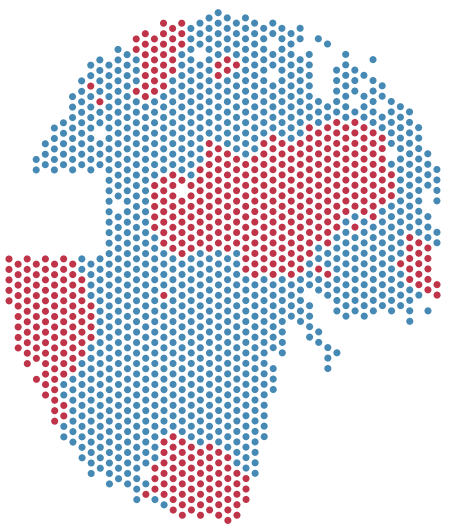
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



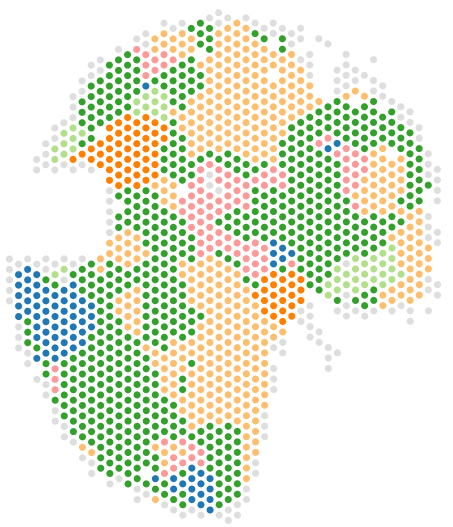
Black line: observed candidate outer edge

Final candidate location



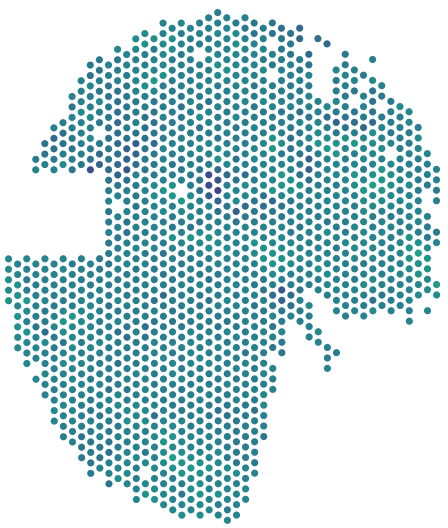
inside outside

Accepted R12 (original 0–8)



1 3 4 5 6 2

Epithelial support



0 1

Fraction of all 30

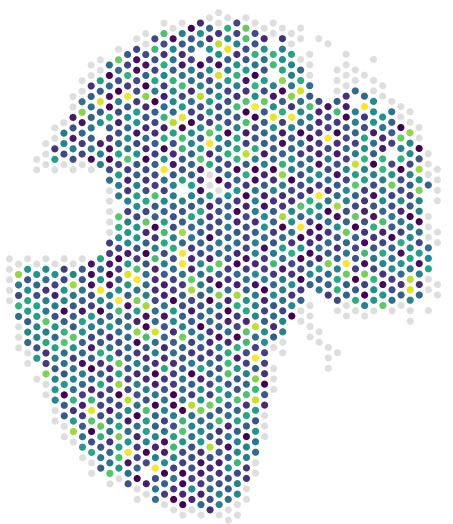
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



0 0.14

Matched-model AUCell

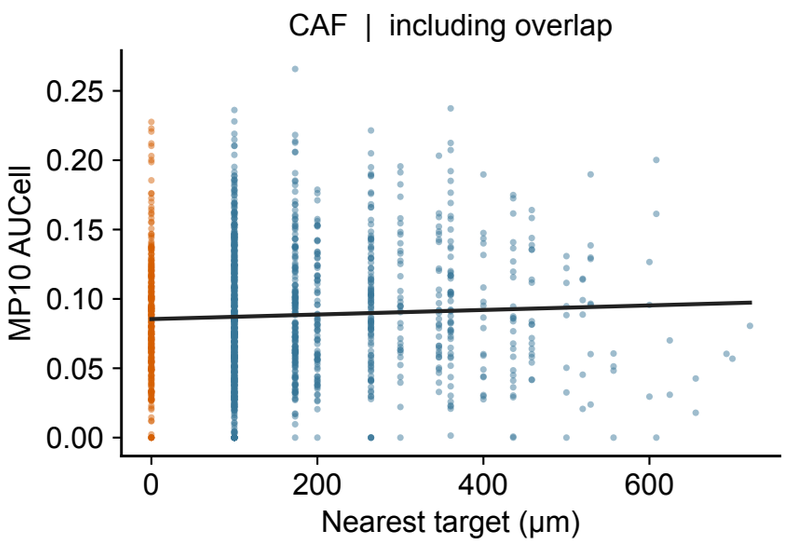
GSE171351_C1 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

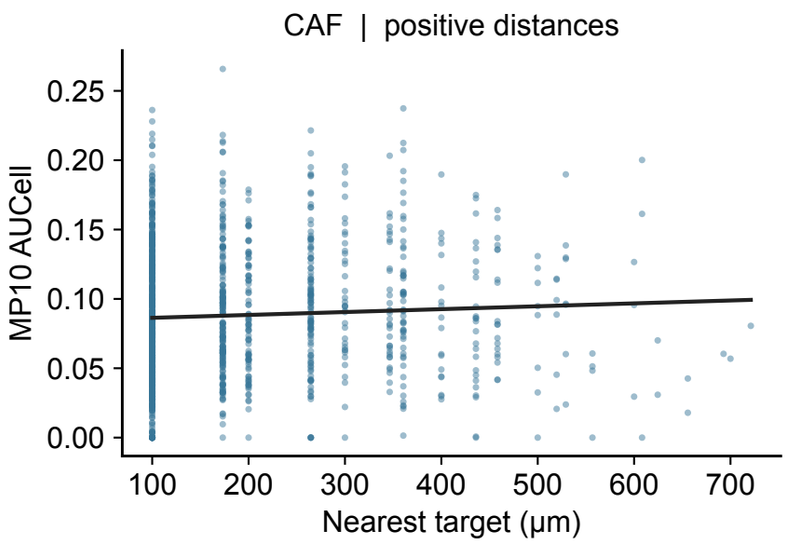


GSE171351_C1 | MP10 versus nearest target

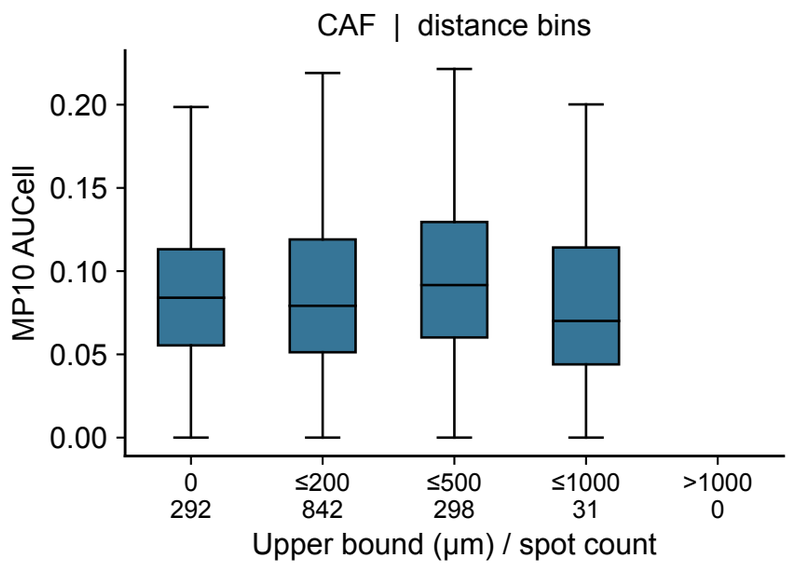
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



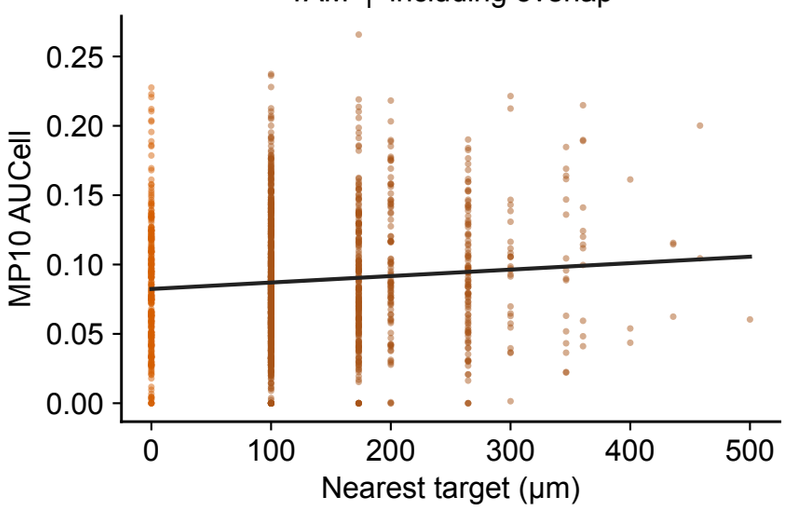
r +0.05 ρ +0.05
n 1463/1465 queries • 294 targets



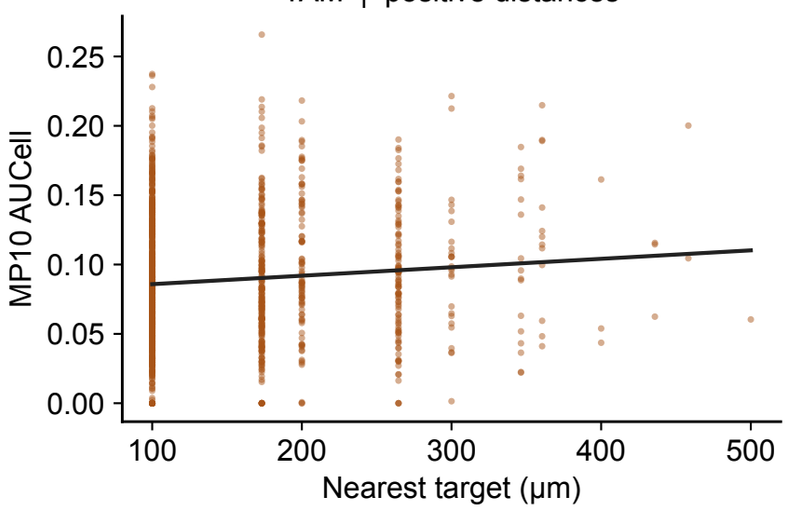
r +0.05 ρ +0.06
n 1171/1465 queries • 294 targets



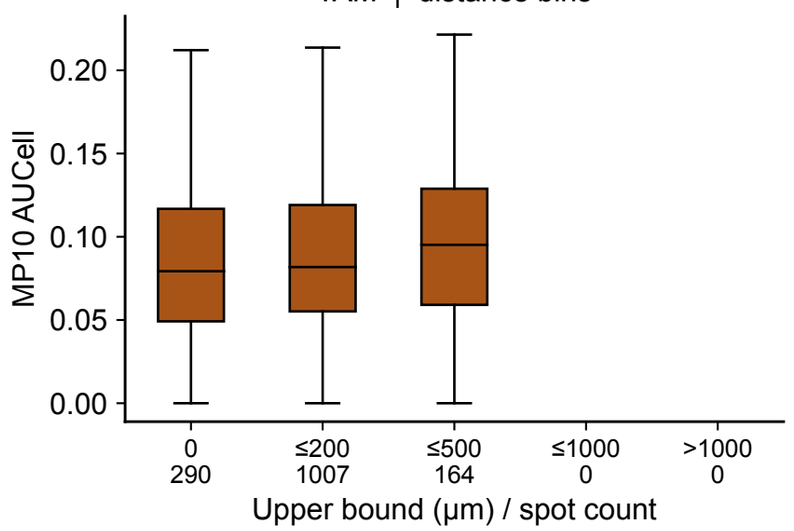
Orange = zero distance
Line = within-section linear fit



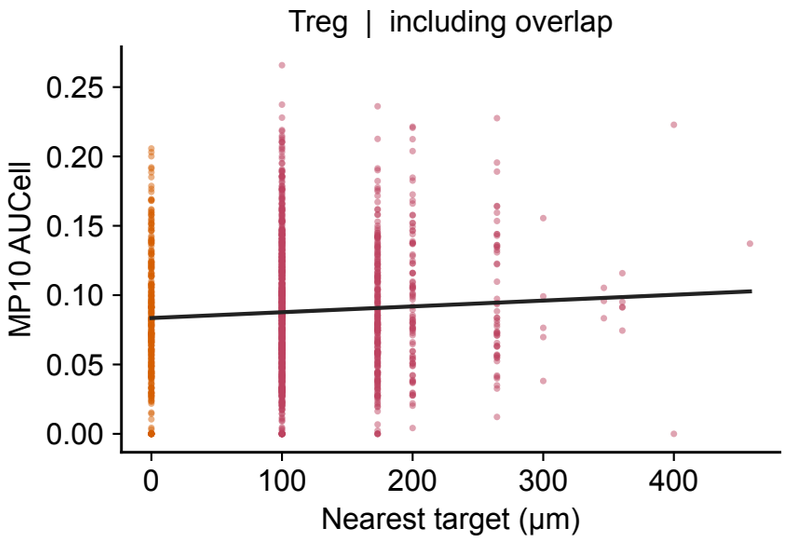
r +0.09 ρ +0.08
n 1461/1465 queries • 294 targets



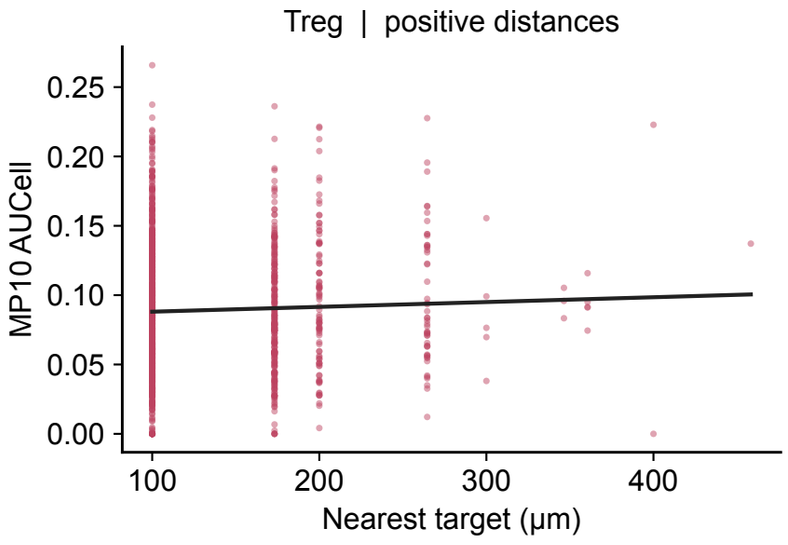
r +0.09 ρ +0.08
n 1171/1465 queries • 294 targets



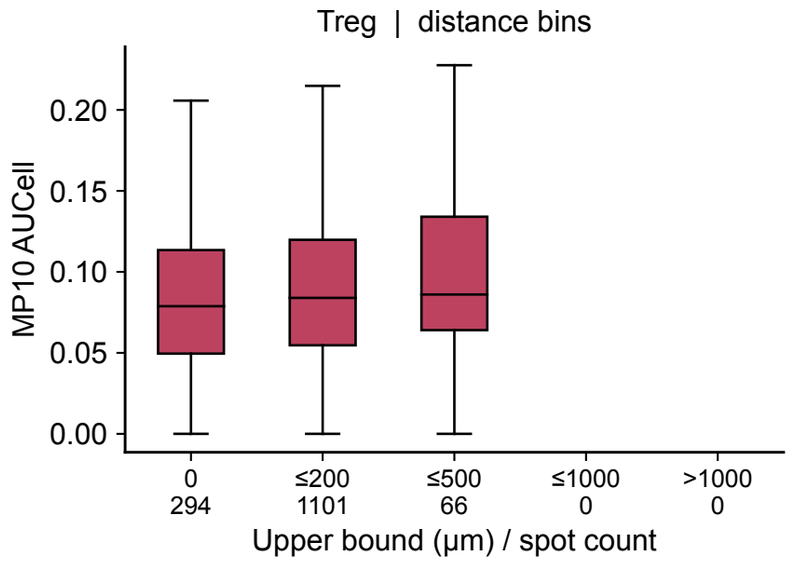
Orange = zero distance
Line = within-section linear fit



r +0.06 ρ +0.05
n 1461/1465 queries • 294 targets



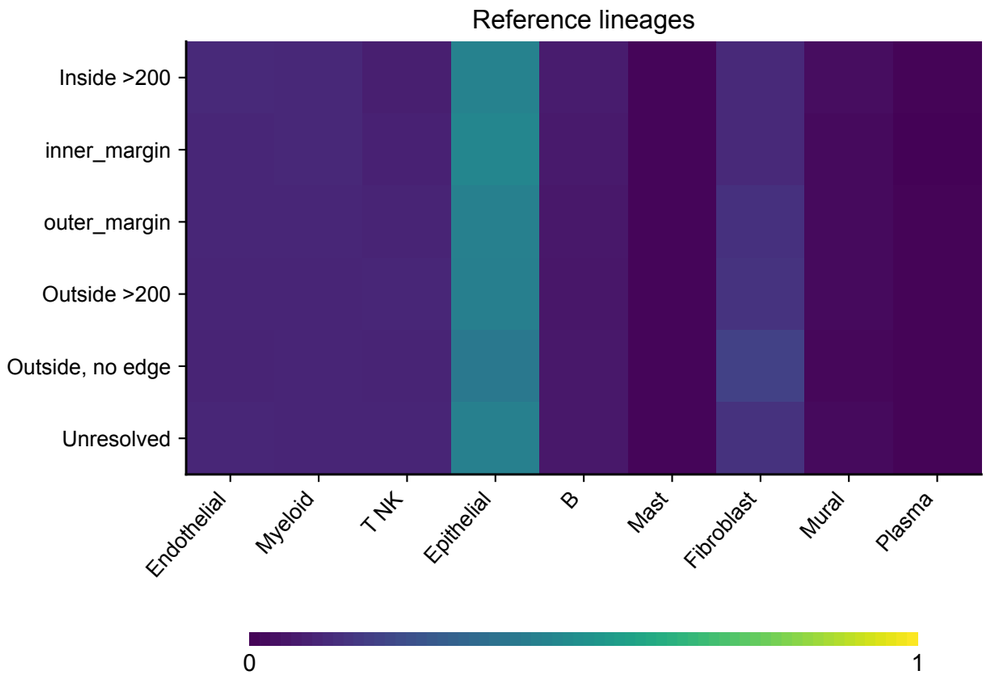
r +0.04 ρ +0.03
n 1167/1465 queries • 294 targets

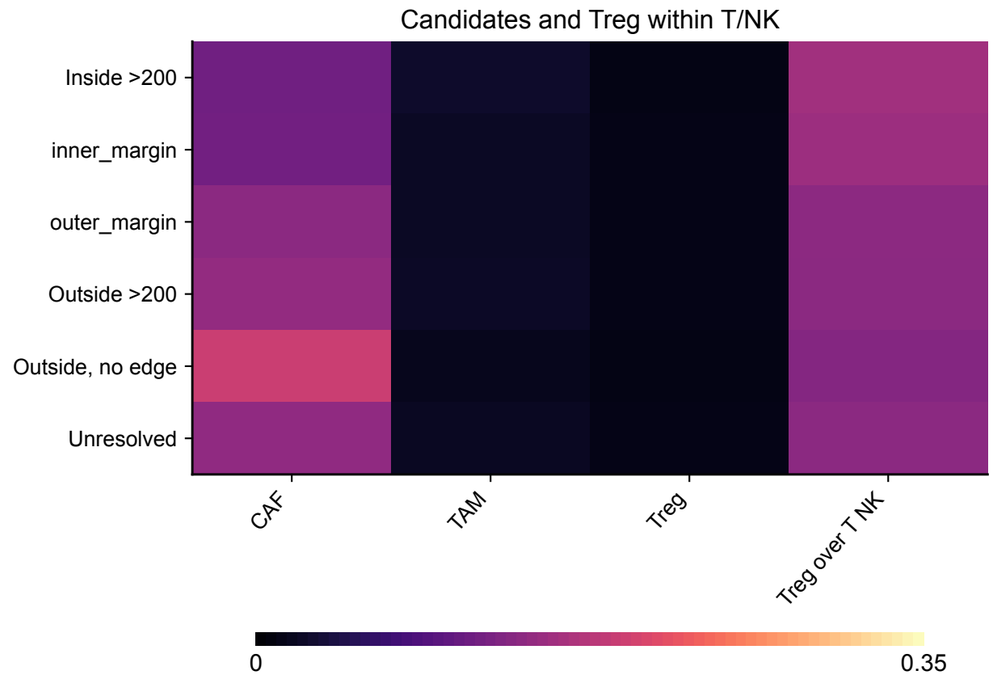


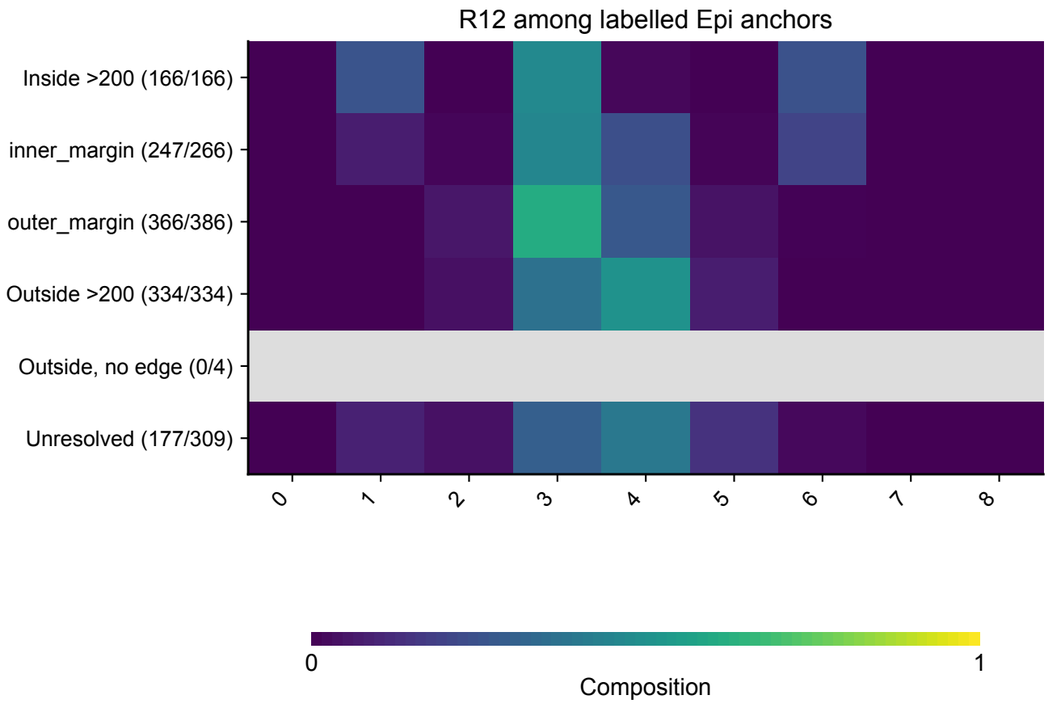
Orange = zero distance
Line = within-section linear fit

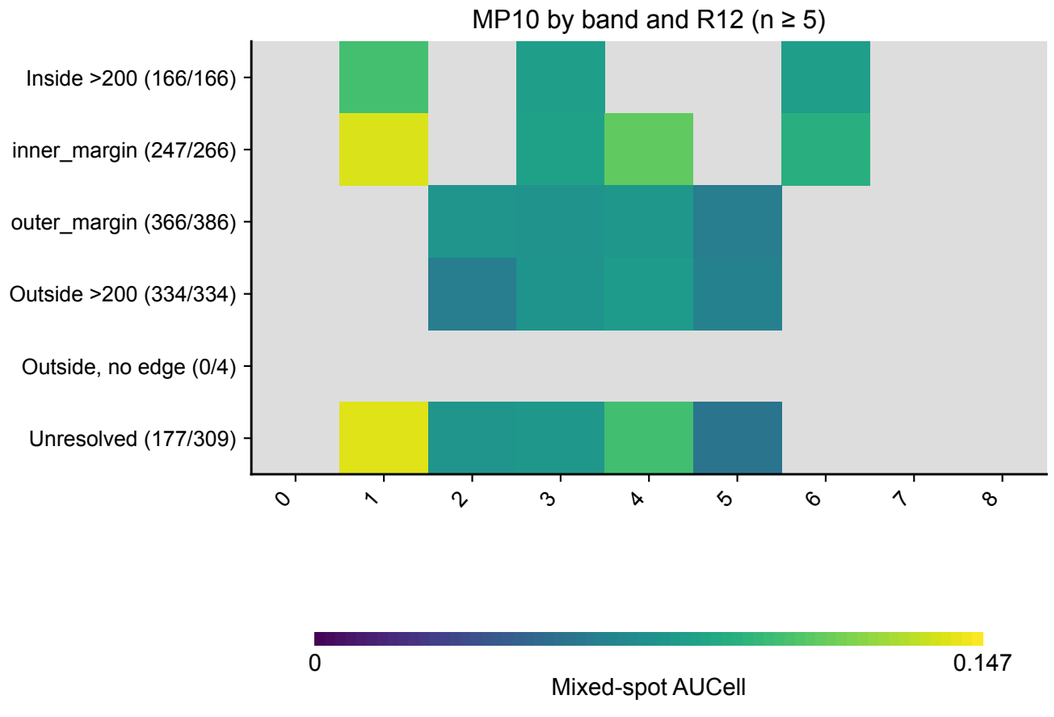
GSE171351_C1 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





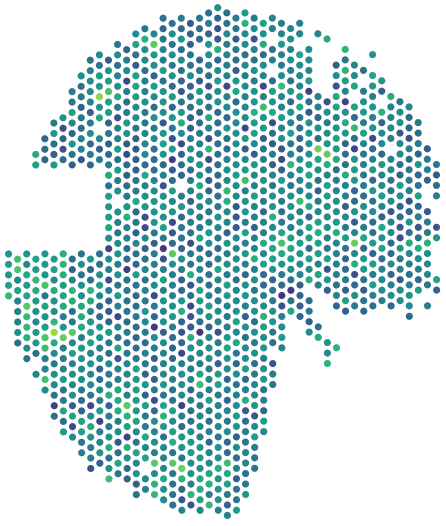




GSE171351_C1 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

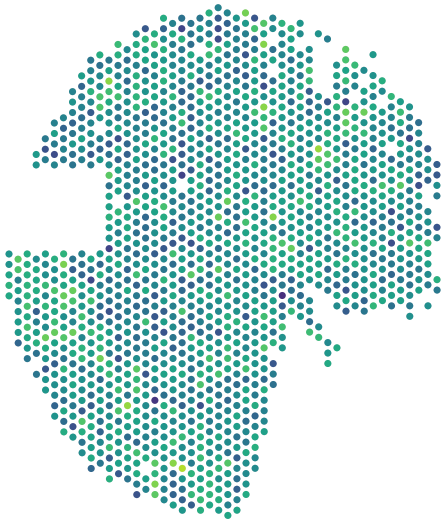
E2F Targets



0 0.152

UCell score

G2M Checkpoint



0 0.134

UCell score

Hypoxia



0 0.211

UCell score

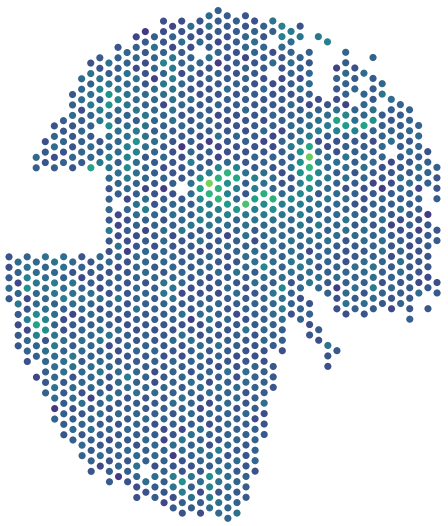
Tnfa Signaling Via Nfkb



0 0.275

UCell score

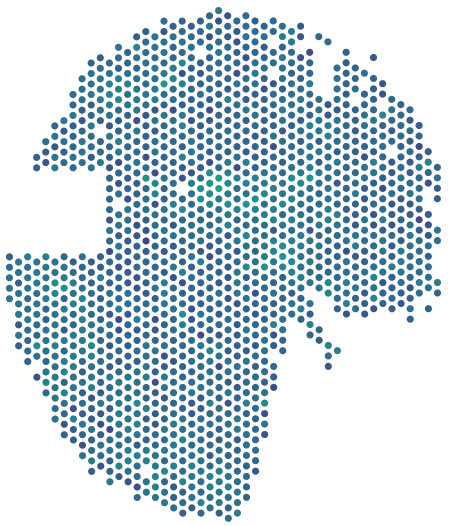
Emt



0 0.386

UCell score

Interferon Gamma Response



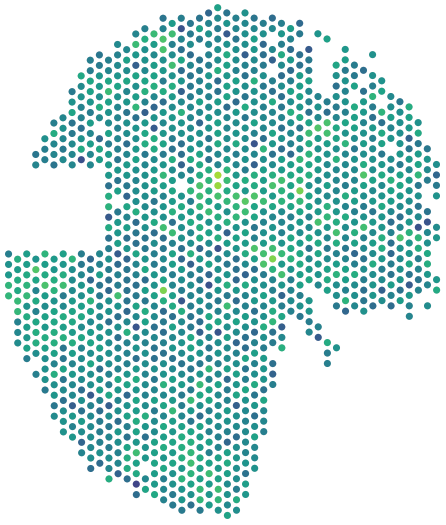
0 0.275

UCell score

GSE171351_C1 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

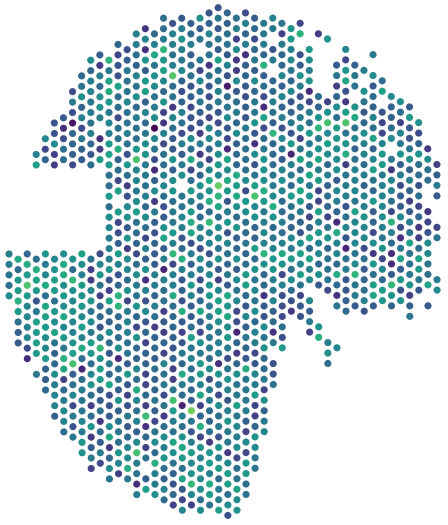
Il2 Stat5 Signaling



0 0.144

UCell score

Tgf Beta Signaling



0 0.232

UCell score

Cholesterol Homeostasis



0 0.21

UCell score

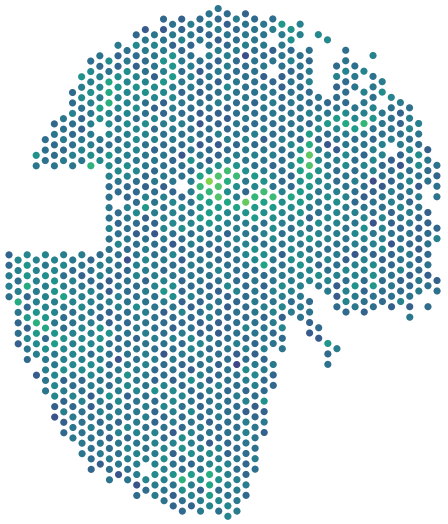
Cholesterol Biosynthesis



0 0.166

UCell score

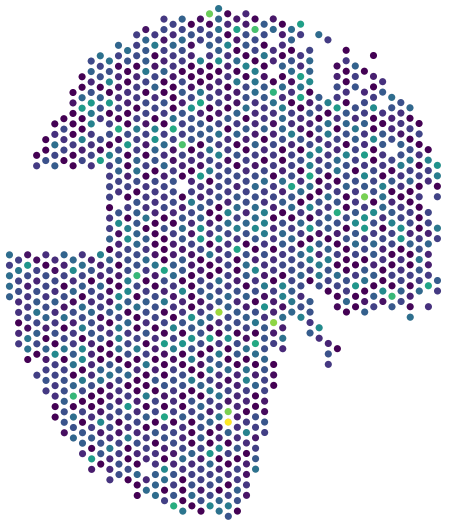
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



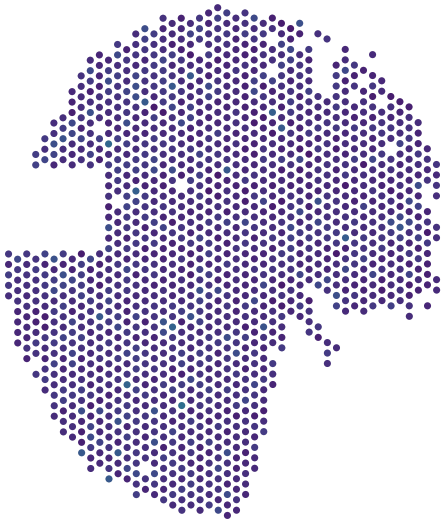
0 0.151

UCell score

GSE171351_C1 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

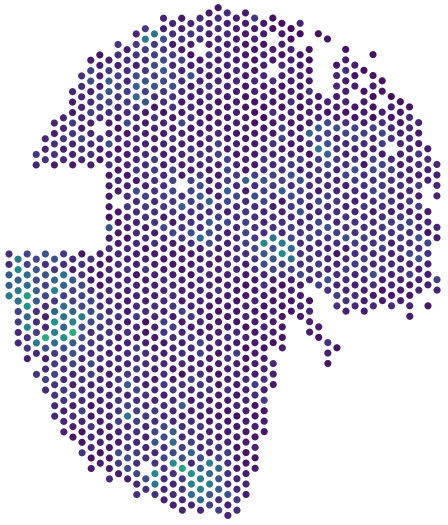
Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn



0 0.0778

UCell score

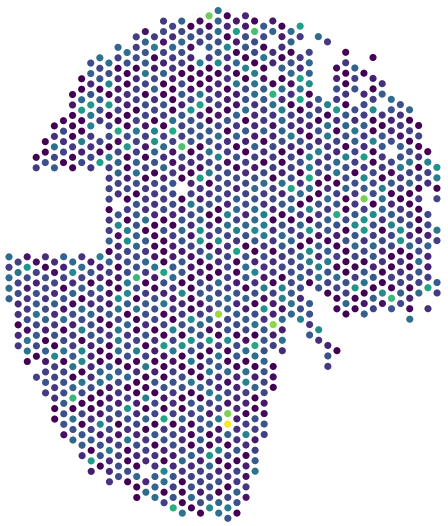
Treg Resting Vs Activated Up



0 0.0609

UCell score

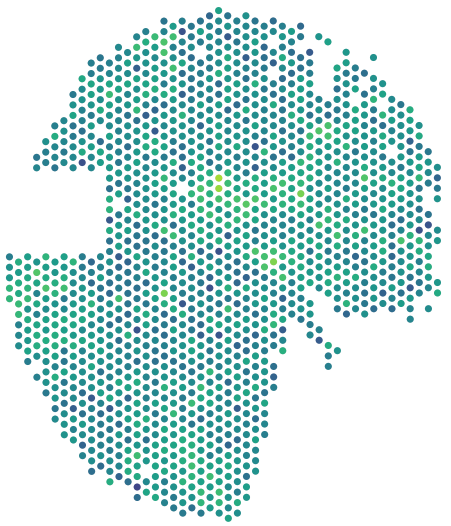
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

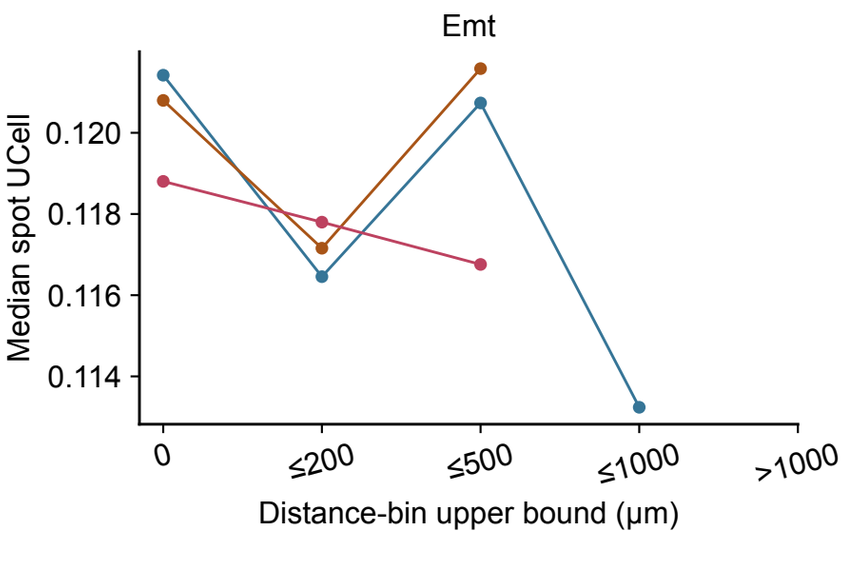


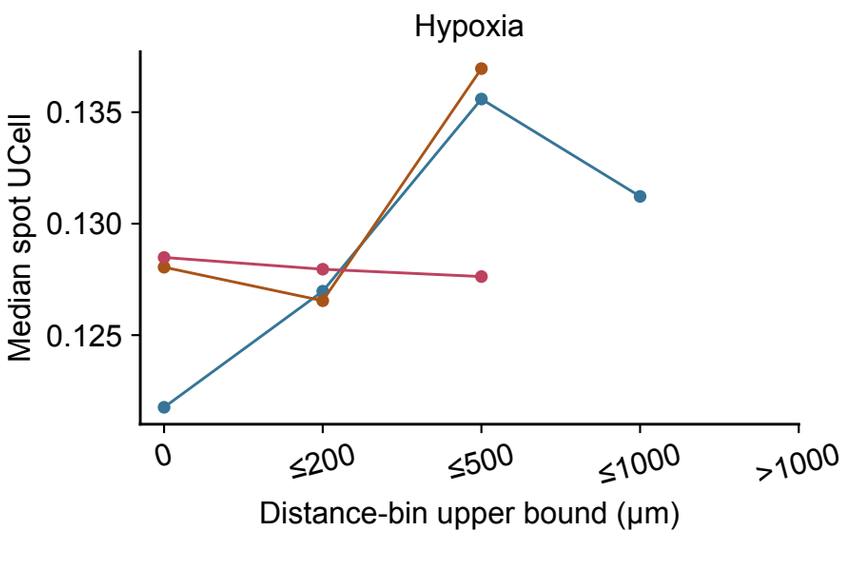
0 0.144

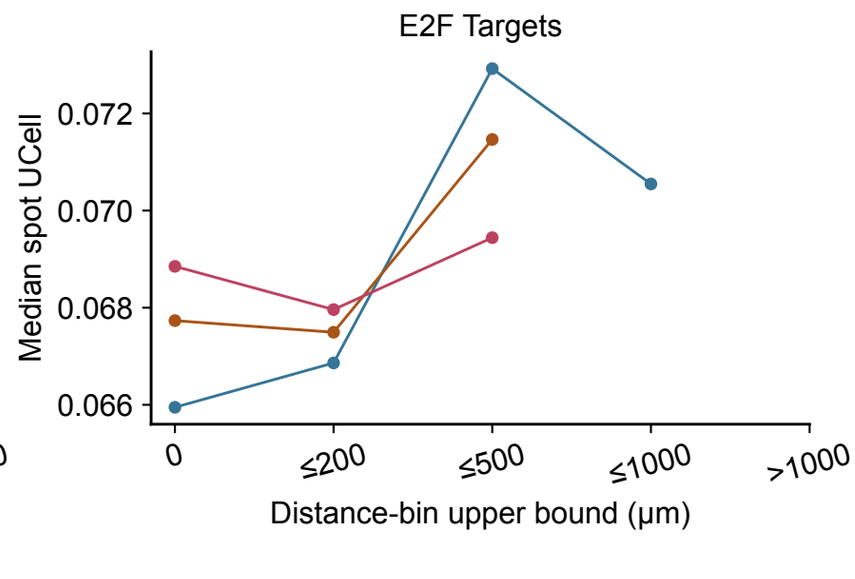
UCell score

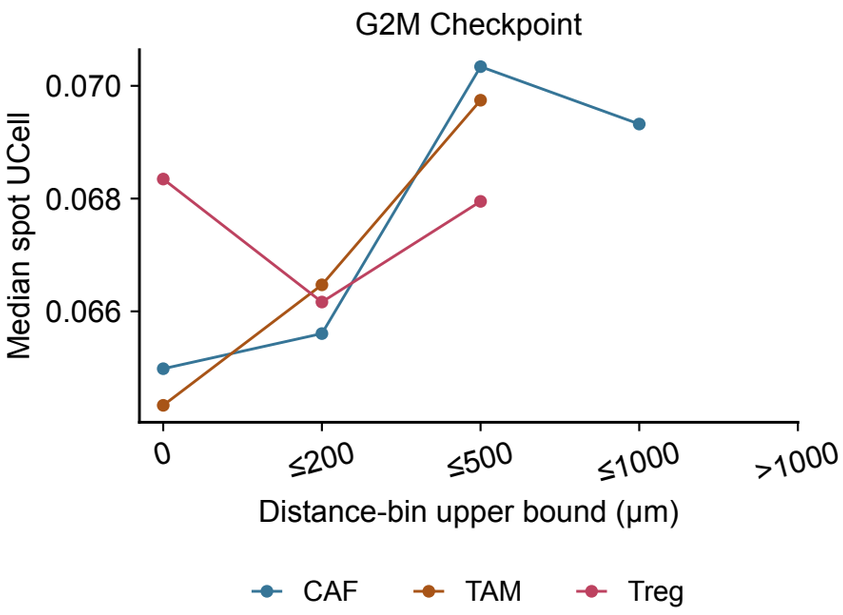
GSE171351_C1 | Programs by target distance

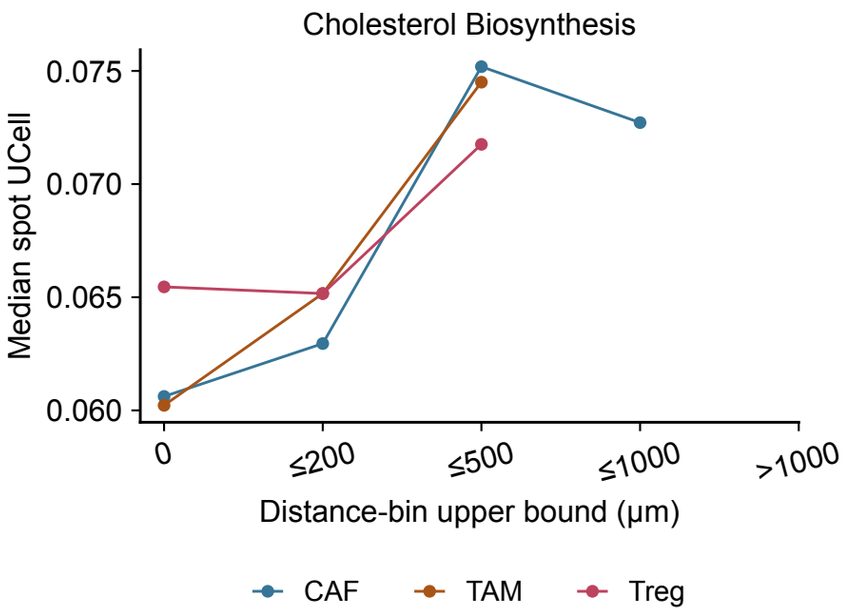
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

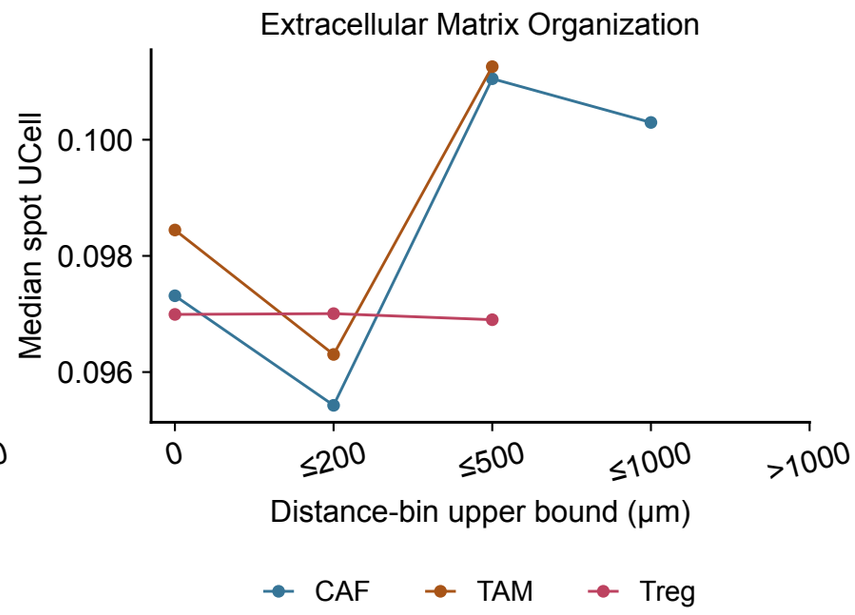






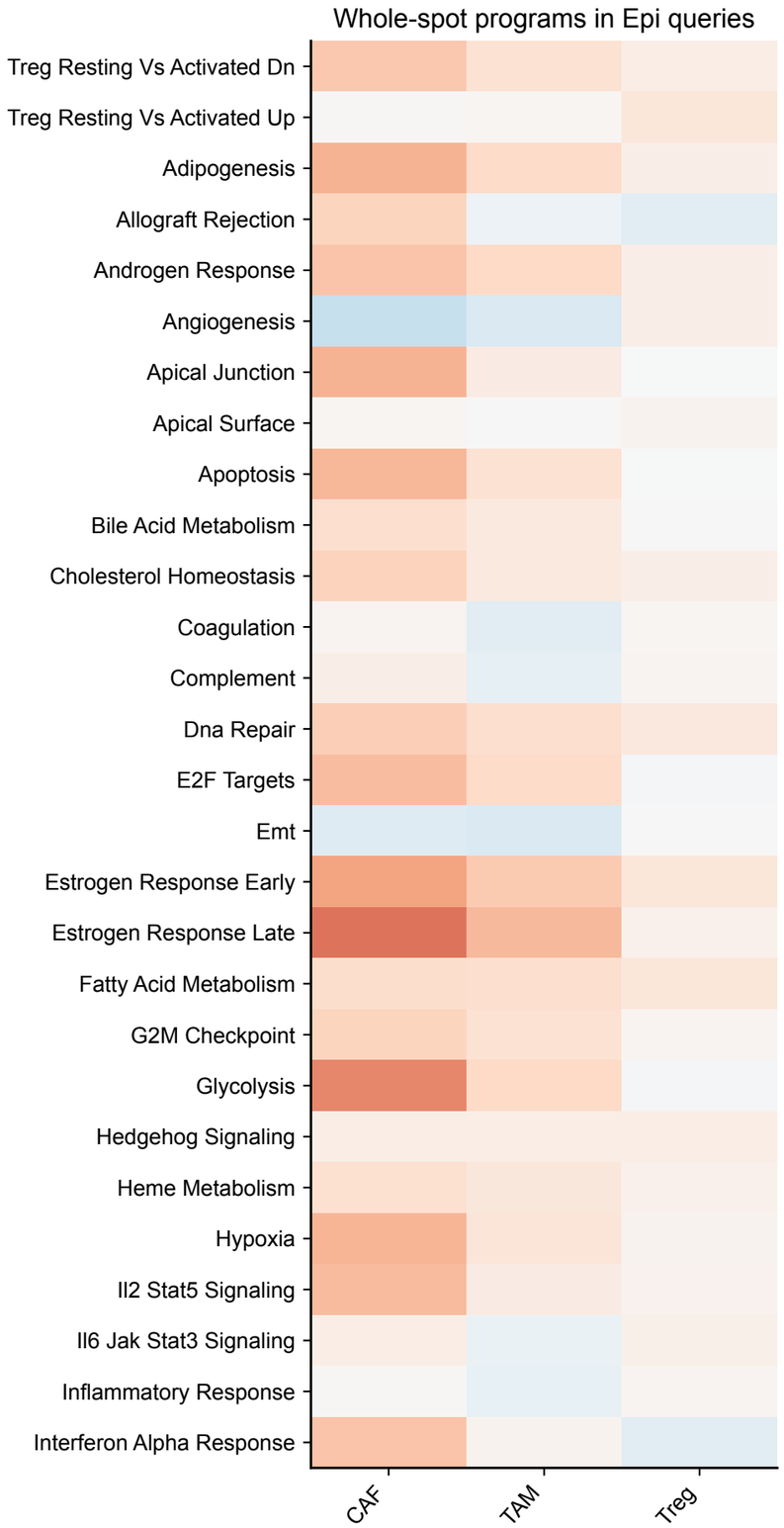


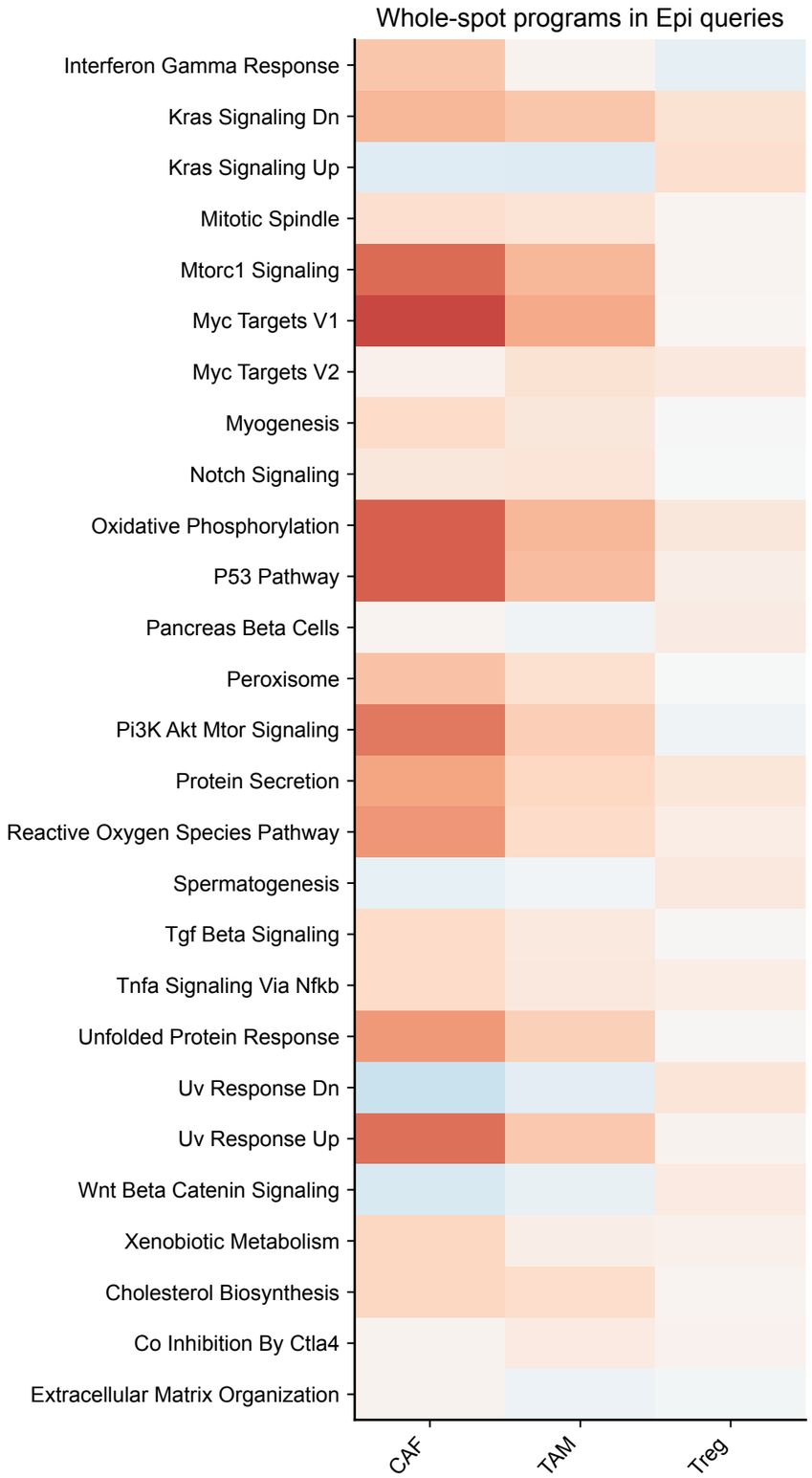




GSE171351_C1 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6





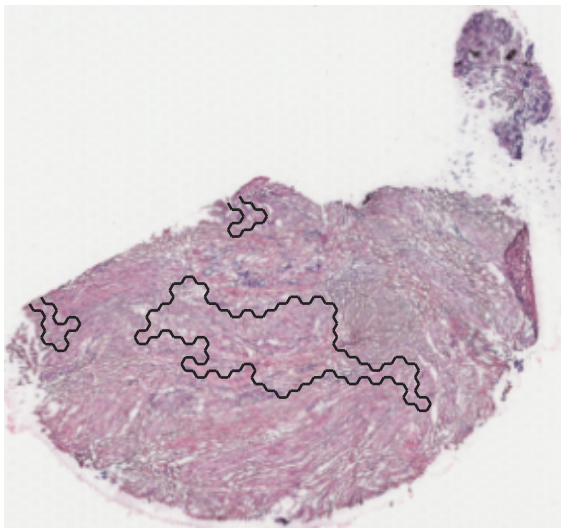




GSE171351_D1 | Tissue context

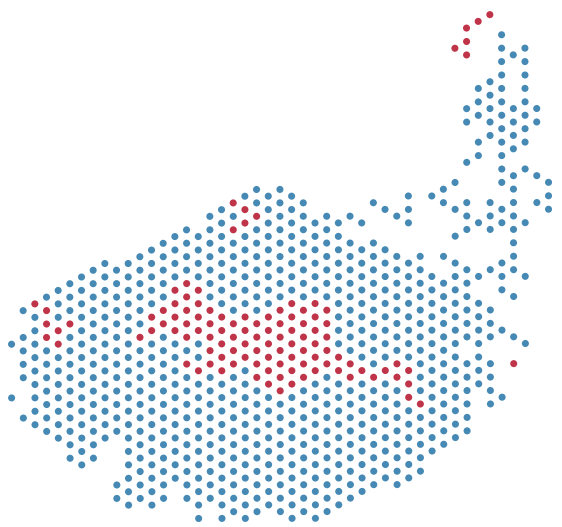
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



Black line: observed candidate outer edge

Final candidate location



Accepted R12 (original 0–8)

No covered labels

Epithelial support

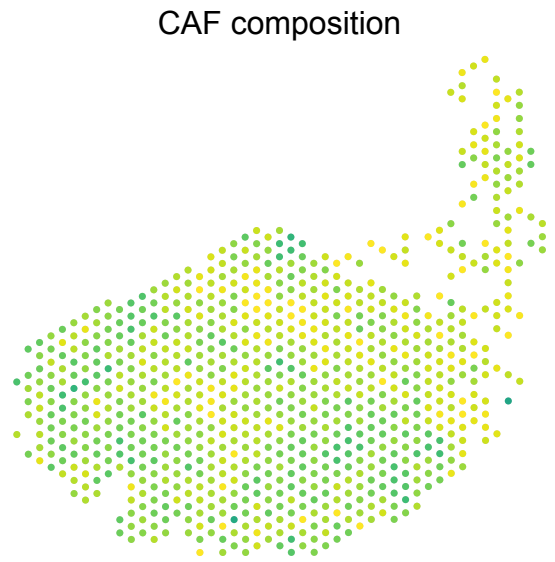


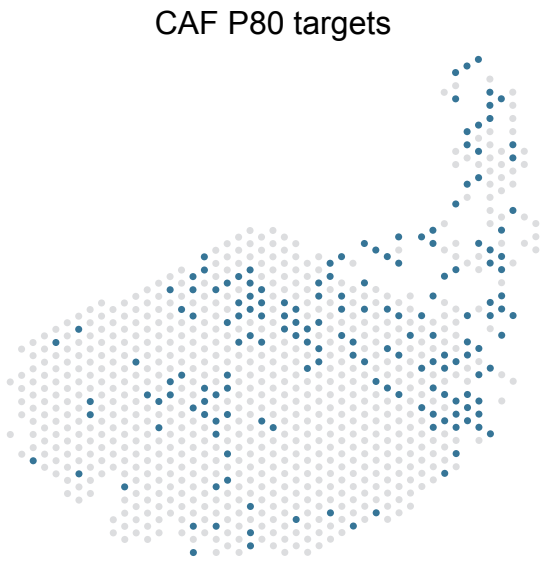
MP10 in Epi queries

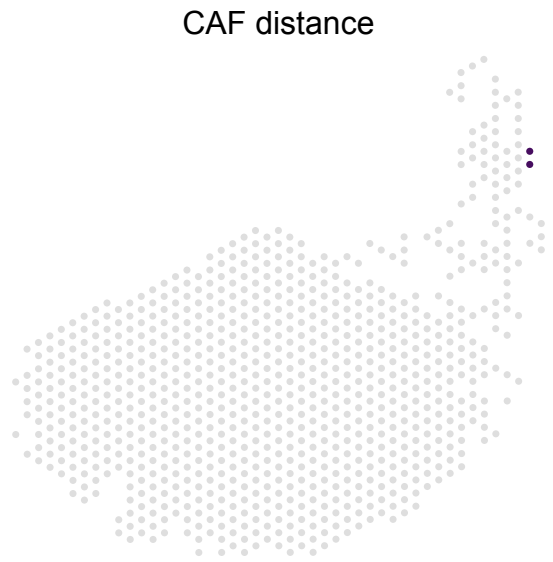
Epi-component MP10 (valid only)

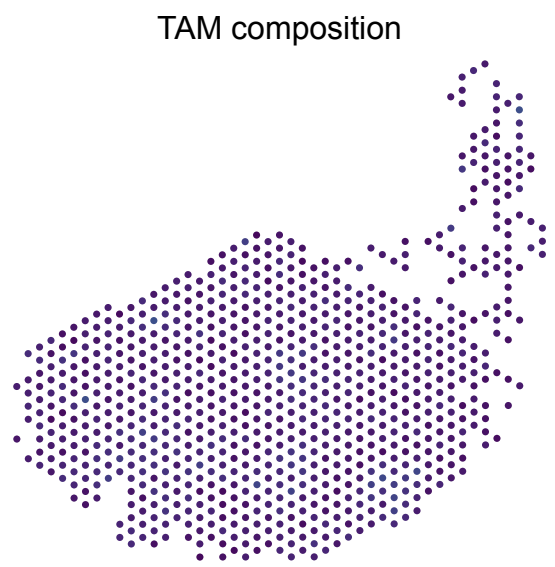
GSE171351_D1 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

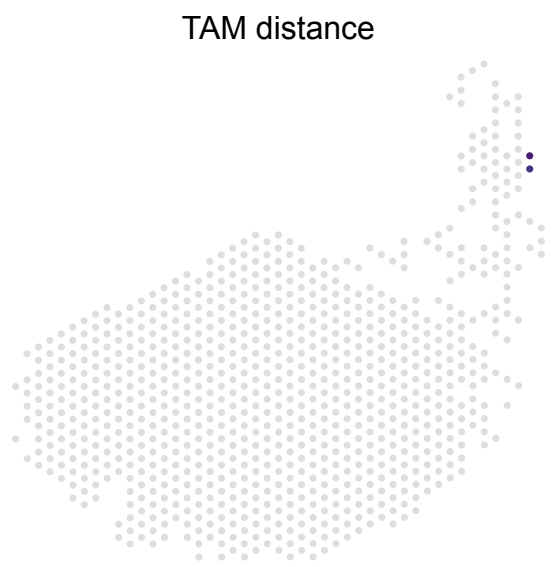


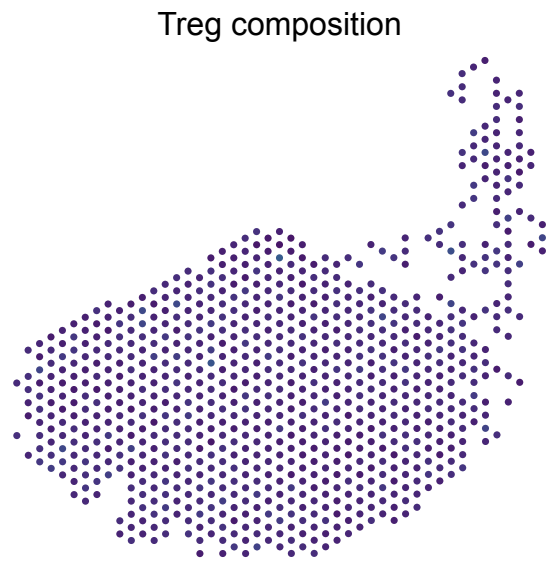


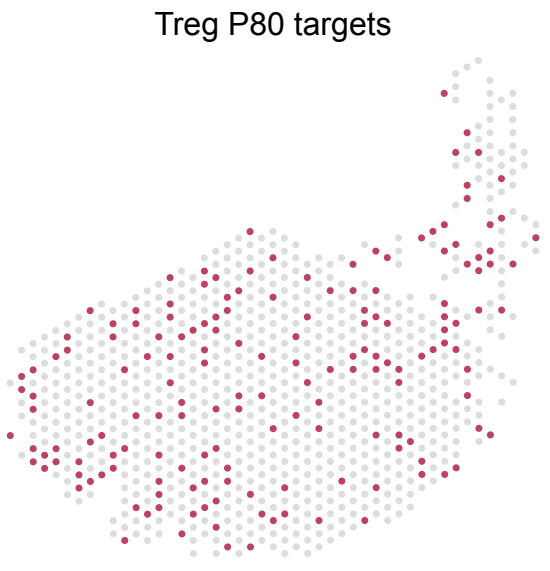








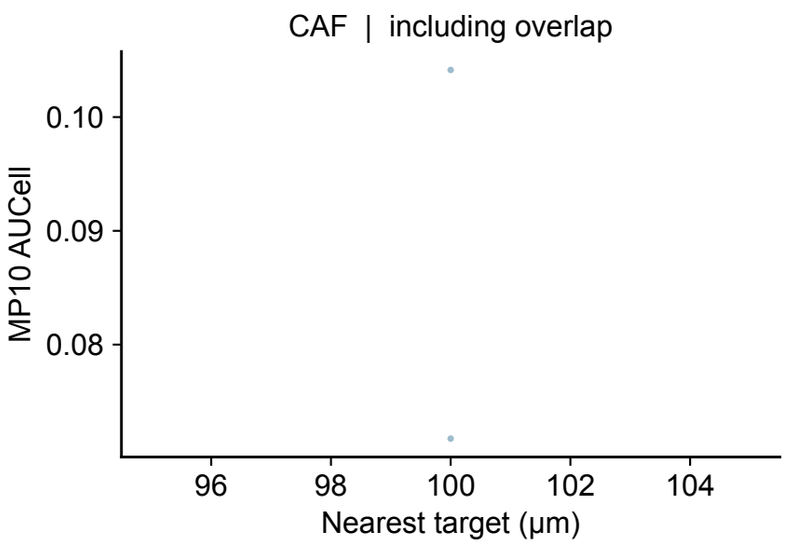




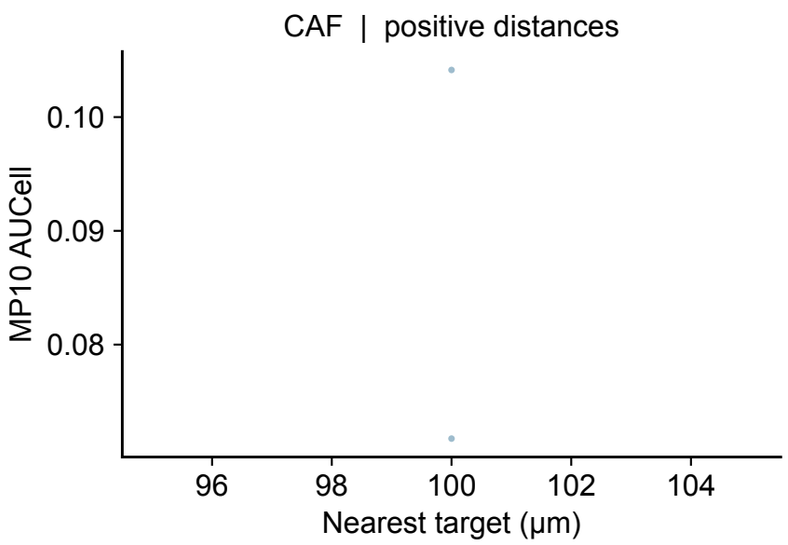


GSE171351_D1 | MP10 versus nearest target

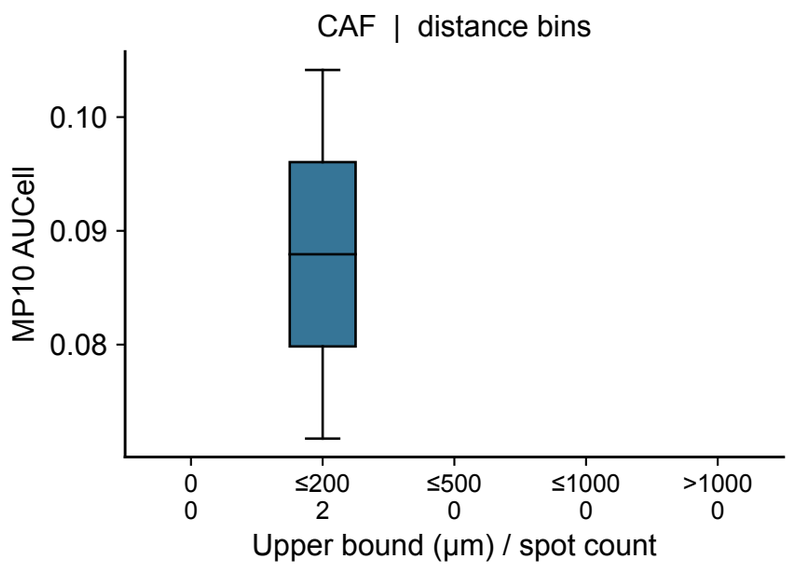
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



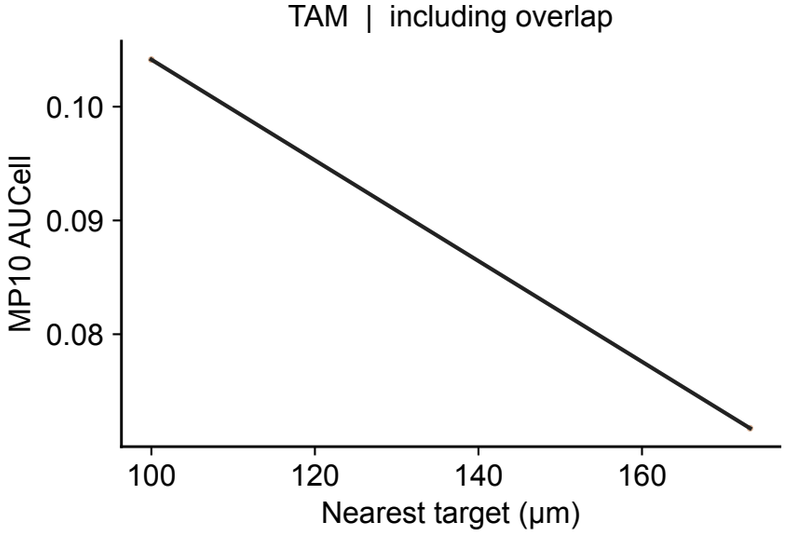
r +nan ρ +nan
n 2/3 queries • 167 targets



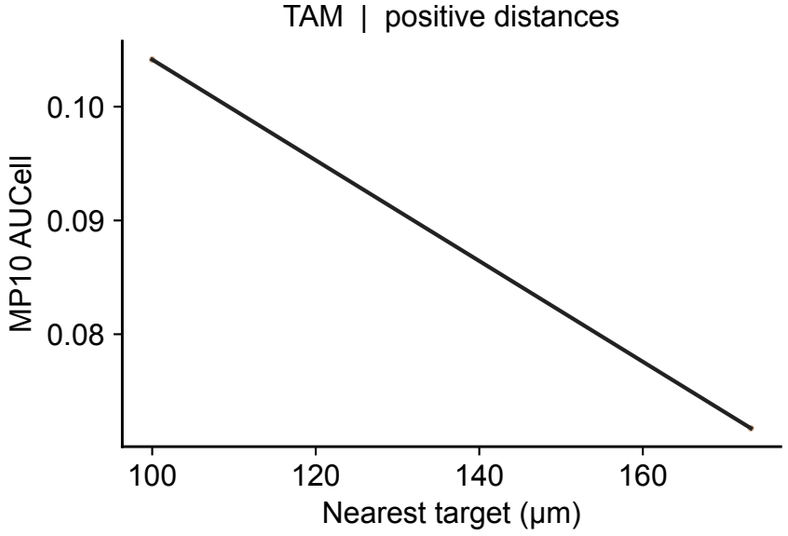
r +nan ρ +nan
n 2/3 queries • 167 targets



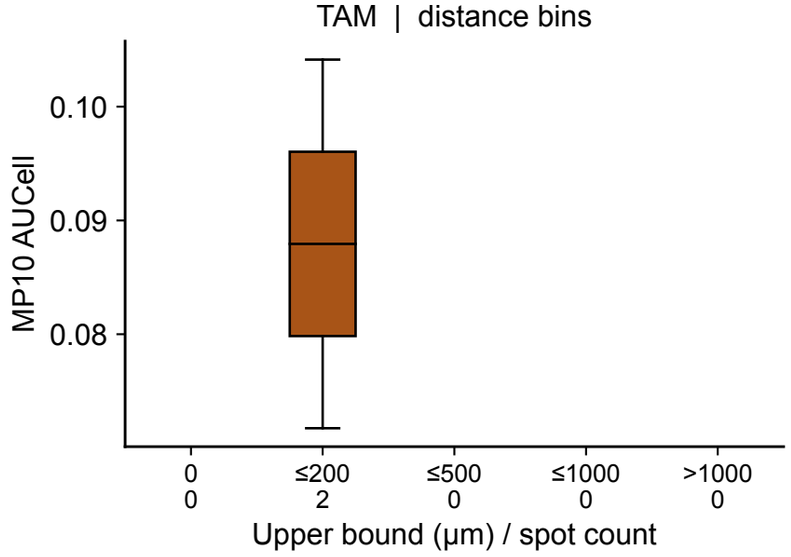
Orange = zero distance
Line = within-section linear fit



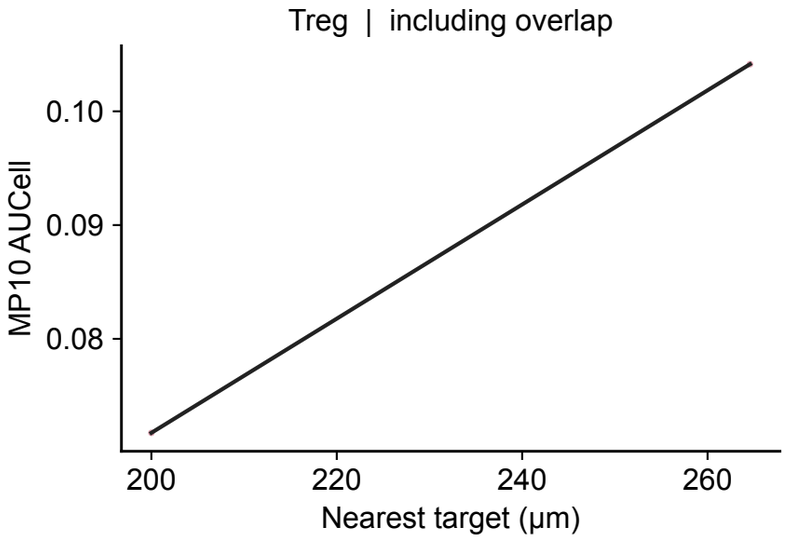
r +nan ρ +nan
n 2/3 queries • 167 targets



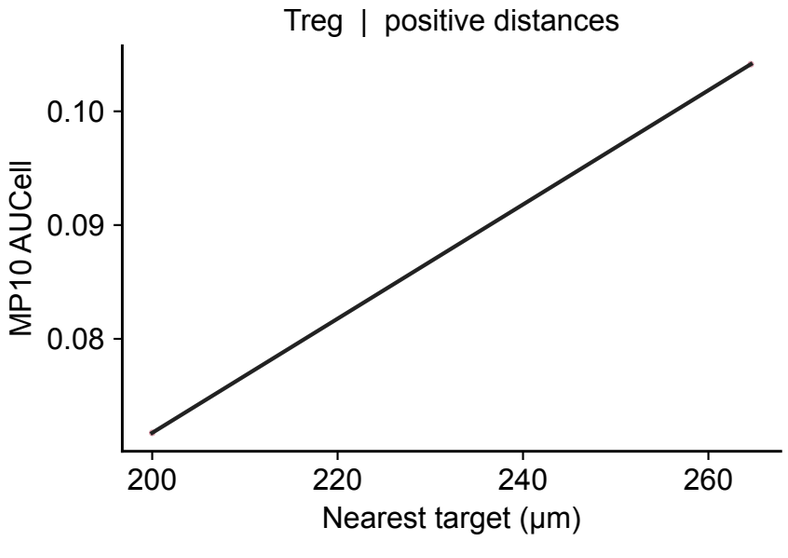
r +nan ρ +nan
n 2/3 queries • 167 targets



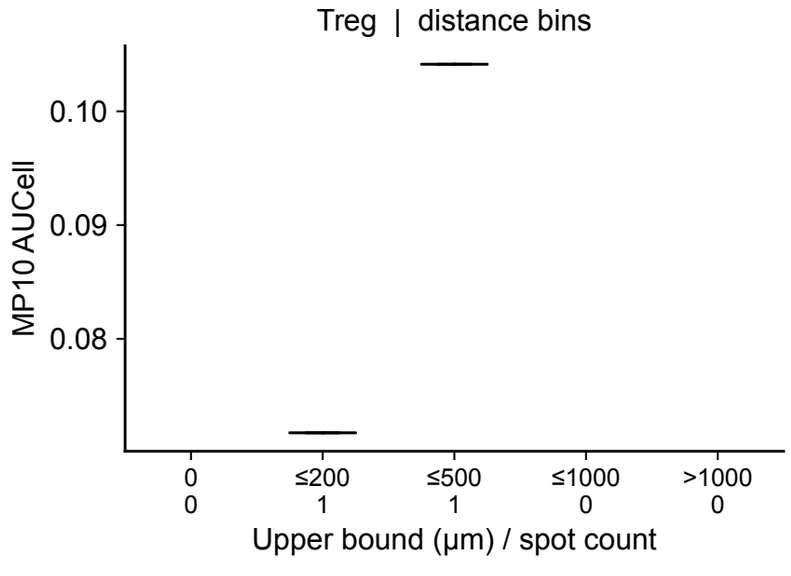
Orange = zero distance
Line = within-section linear fit



r +nan ρ +nan
n 2/3 queries • 167 targets



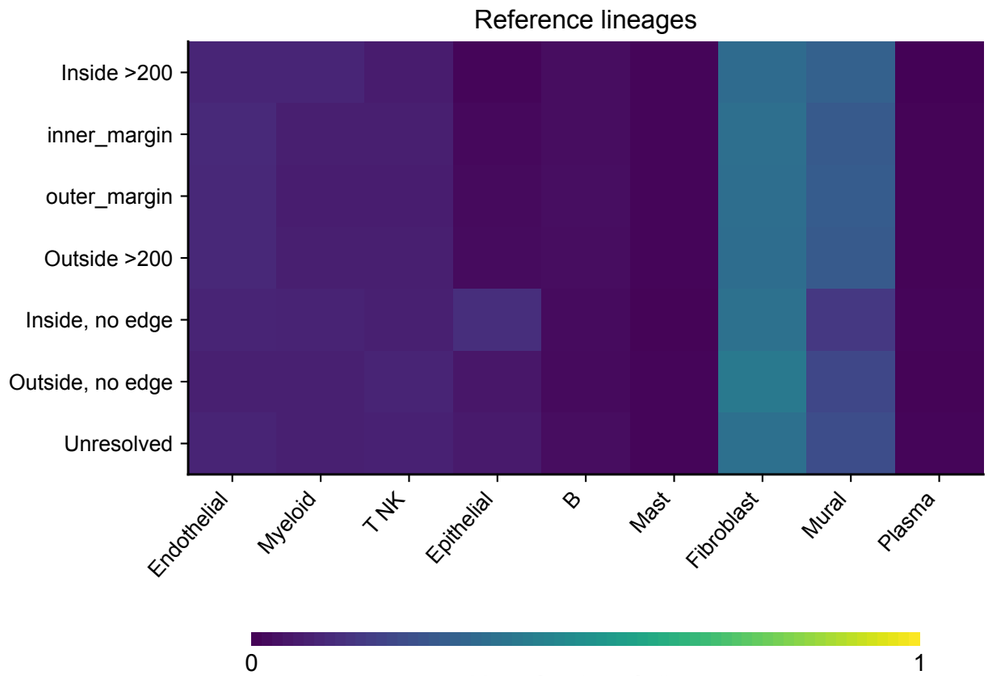
r +nan ρ +nan
n 2/3 queries • 167 targets

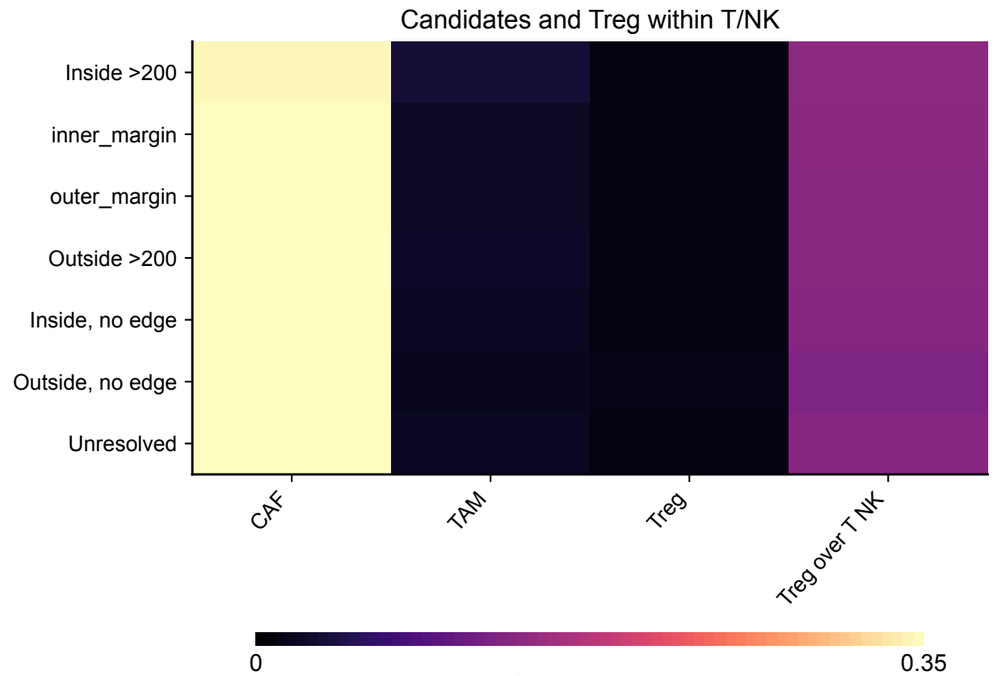


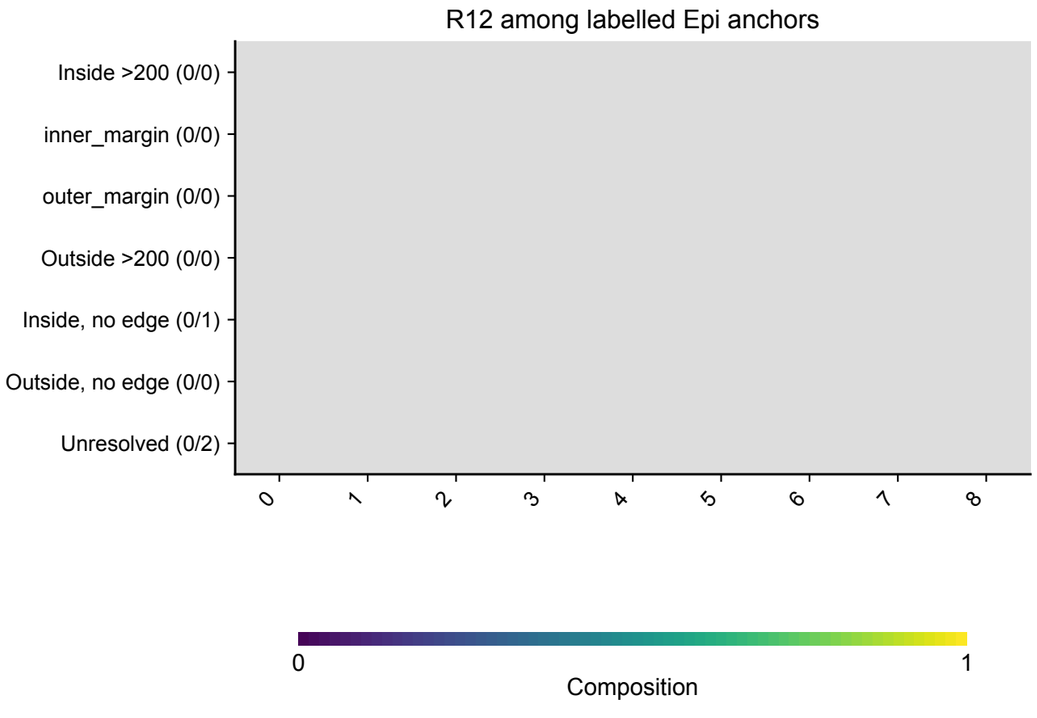
Orange = zero distance
Line = within-section linear fit

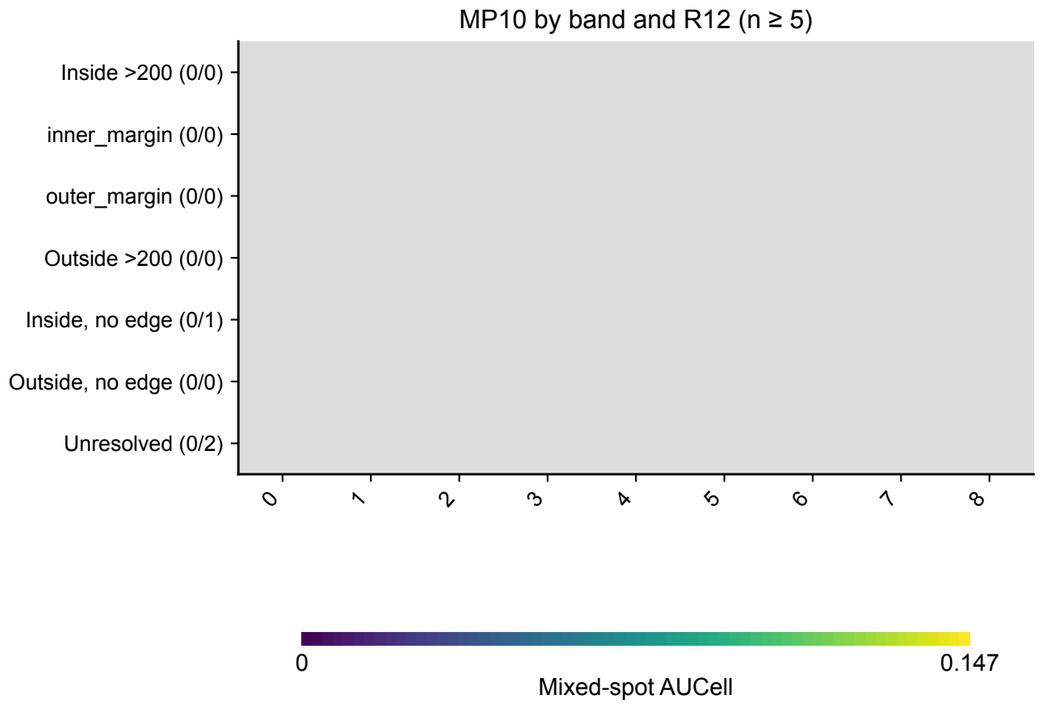
GSE171351_D1 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable



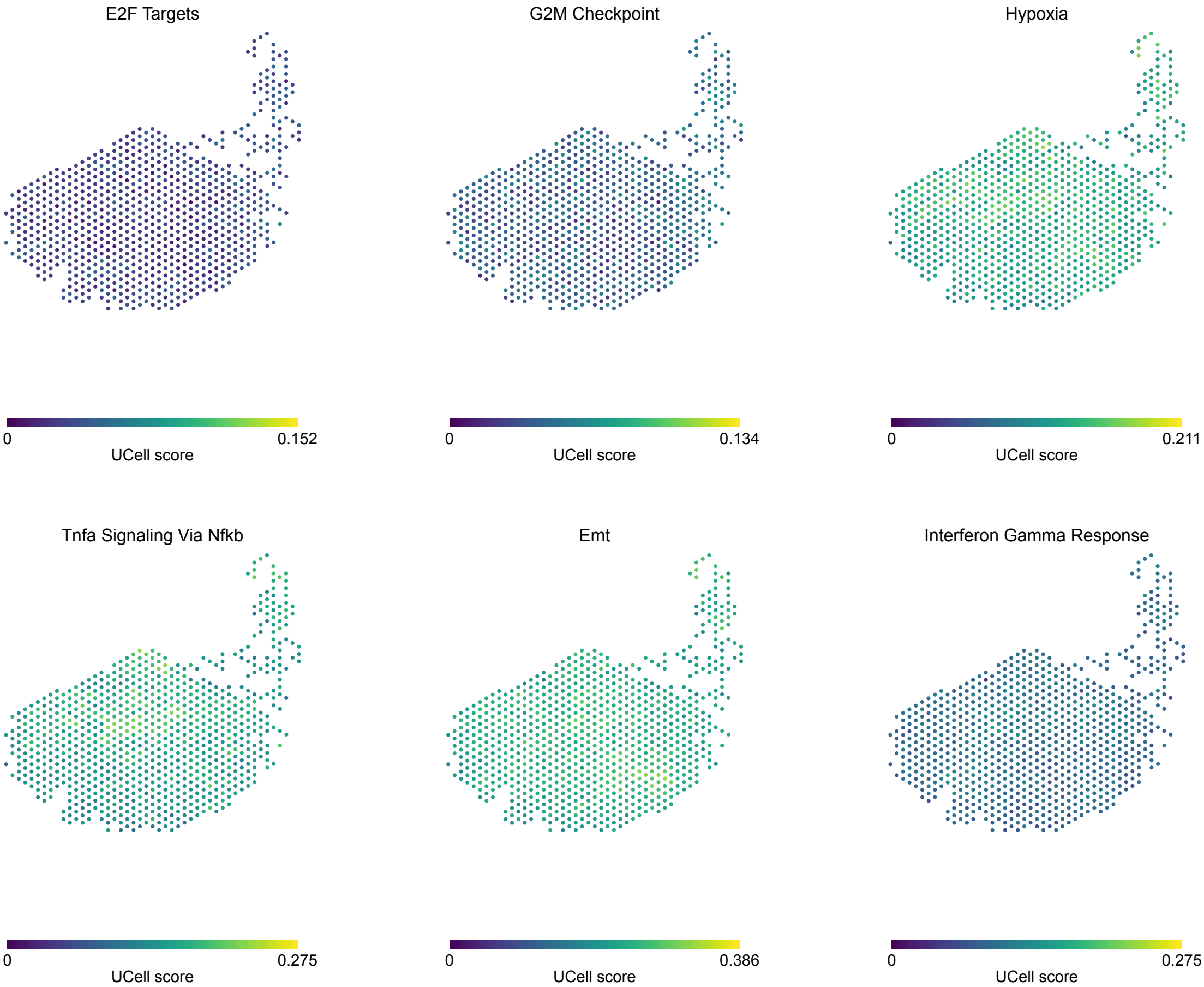






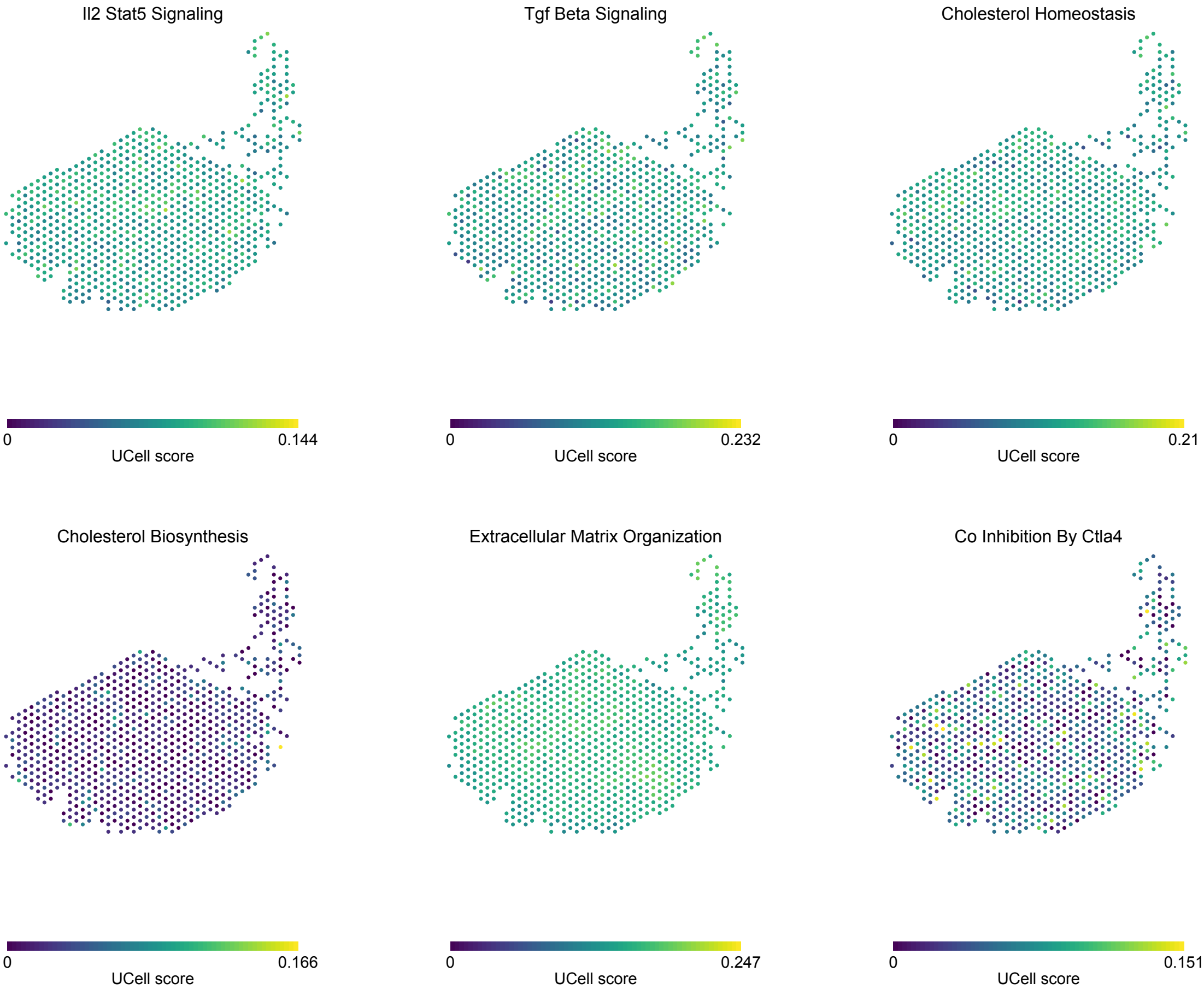
GSE171351_D1 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections



GSE171351_D1 | Immune, lipid and matrix programs

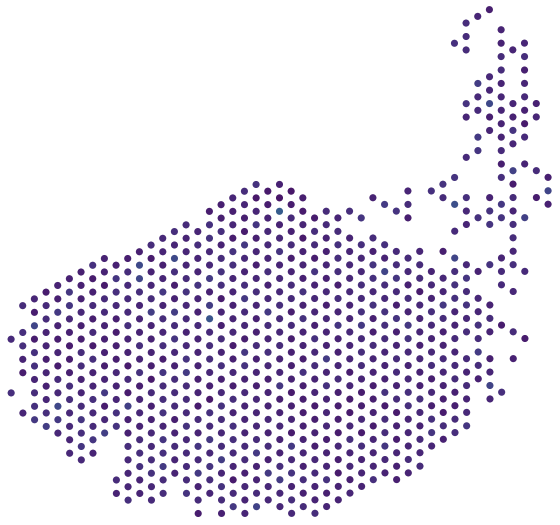
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections



GSE171351_D1 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn



0 0.0778

UCell score

Treg Resting Vs Activated Up



0 0.0609

UCell score

Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

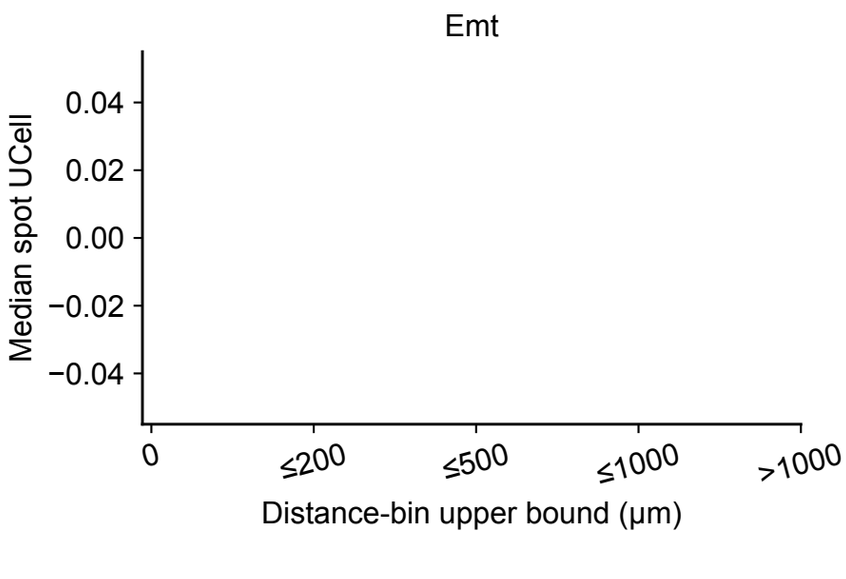


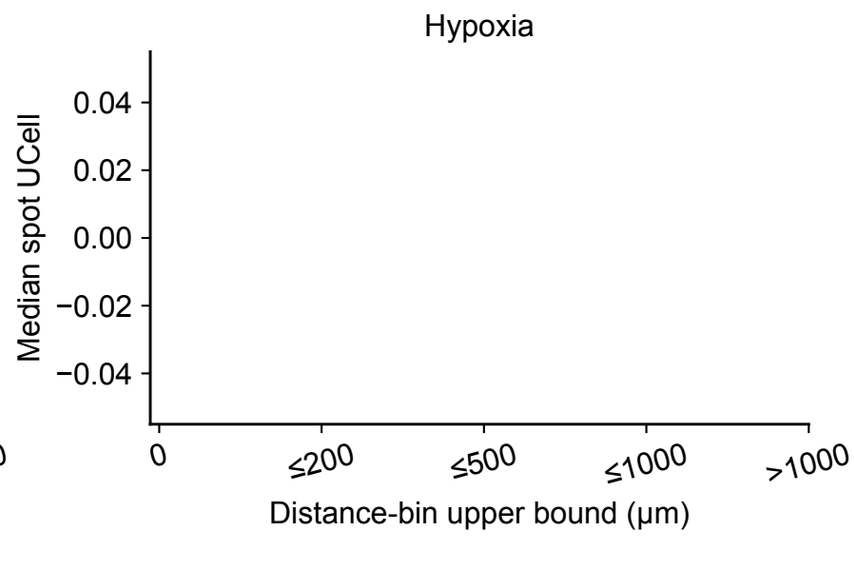
0 0.144

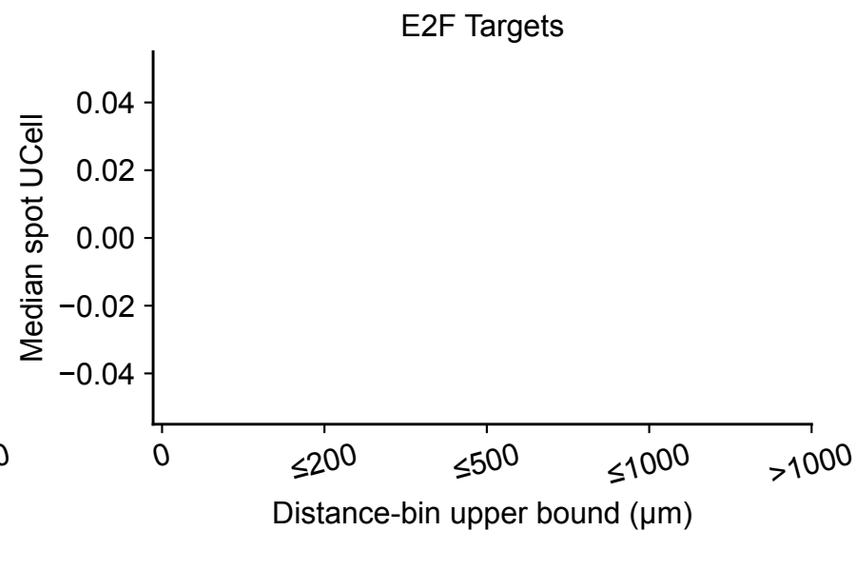
UCell score

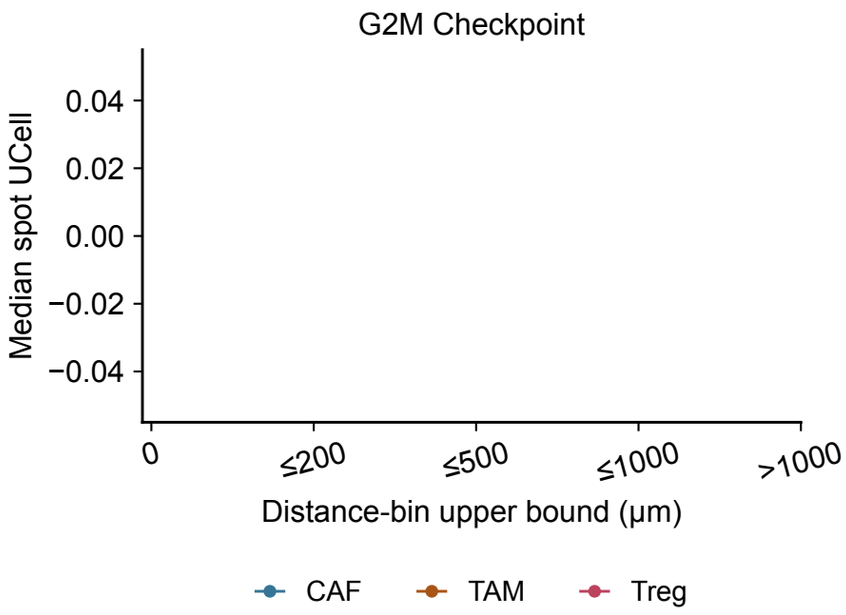
GSE171351_D1 | Programs by target distance

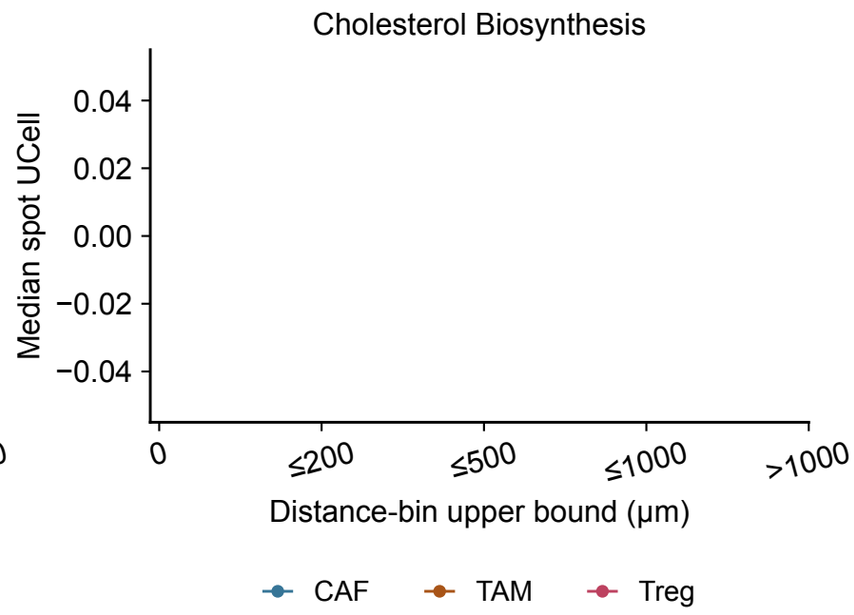
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

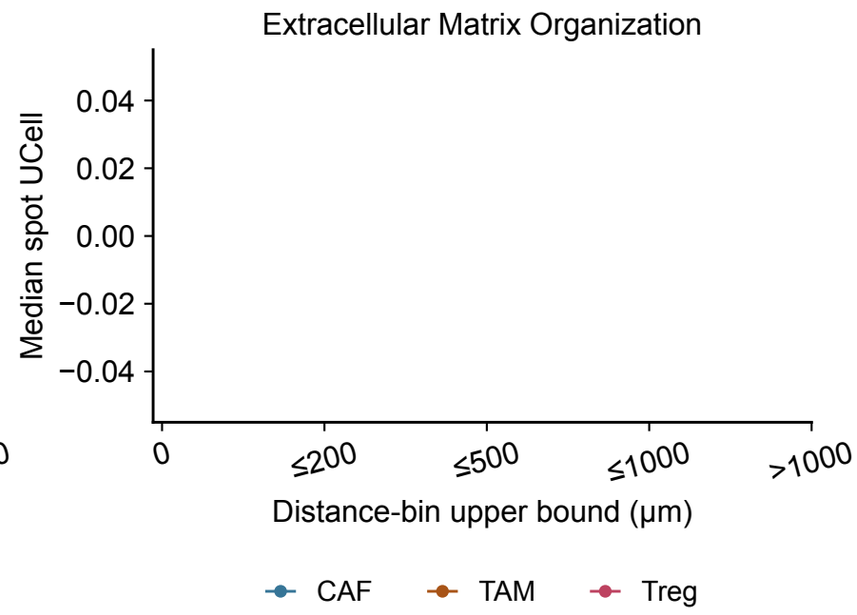












GSE171351_D1 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

